

Modeling News-Driven Liquidity Dynamics and Information Propagation in Commodities Markets

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Glossary

This glossary defines terms from finance, statistics, and machine learning that recur in this thesis, in plain language, for readers who are not specialists in these fields. Terms are listed alphabetically.

Ablation study. Removing or changing one component of a model at a time to measure how much that component matters, like testing a recipe by leaving out one ingredient at a time.

Amihud illiquidity ratio. A measure of how much the price moves for a given amount of trading. A high value means even small trades push the price a lot, so the market is illiquid.

Asymmetry (directional). When two opposite cases produce different-sized effects. Here, whether negative news moves the market more than positive news of the same strength.

Bearish / Bullish. Bearish means expecting the price to fall (news that is bad for the price). Bullish means expecting it to rise (news that is good for the price).

Channel (supply, demand, risk premium). In this thesis, one of three separate economic dimensions a news item can affect: how much oil is available (supply), how much is consumed (demand), or how much geopolitical or operational risk it implies (risk premium).

Chokepoint. A narrow, critical passage in a supply route (for example the Strait of Hormuz) whose disruption can cut off a large share of the world's oil flow.

Coefficient. In a regression, a number that says how much the outcome changes when a given input increases by one unit.

Contemporaneous. Happening at the same time. A contemporaneous effect is felt in the same hour the news arrives, with no delay.

Cross-sectional. An analysis that treats each observation (here, each article) on its own, rather than as a connected sequence over time.

DXY (U.S. Dollar Index). A measure of the dollar's value against a basket of major currencies. Oil is priced in dollars, so it is a relevant control variable.

EIA (Energy Information Administration). The U.S. agency that publishes weekly crude oil inventory figures, a regularly scheduled and closely watched event in oil markets.

Embedding. A way of turning a category (such as “supply” versus “demand”) into a small list of numbers the model can learn from, instead of an arbitrary code.

Encoder / Decoder. In a forecasting model, the encoder reads the past window of data and the decoder produces the forecast for the future steps.

Estimand. The specific quantity a method is actually estimating. Two methods can both study “asymmetry” yet target different estimands, so their answers need not agree.

Front-month futures. The futures contract closest to expiry, which is the most actively traded one for a commodity like crude oil.

Futures contract. An agreement to buy or sell an asset at a set price on a future date. It is the main way crude oil is traded.

Lag. A delay. A “+6 hour lag” effect is one that appears six hours after the news, not immediately.

Liquidity. How easily an asset can be traded without moving its price. It is measured here mainly by hourly trading volume.

Logarithm (log), log volume. A mathematical transform that compresses large numbers. Taking the log of trading volume makes its large swings easier to model.

Marginal effect. The change in the outcome from a one-unit increase in one input, holding everything else fixed.

Multi-horizon forecast. A forecast that predicts several steps ahead at once (here, 1, 3, 6, and 12 hours), rather than only the next step.

OLS (ordinary least squares). The standard method for fitting a straight-line (linear) relationship between inputs and an outcome.

OPEC / OPEC+. The group of major oil-producing countries whose production decisions strongly affect global oil supply and price.

Orthogonal. Independent and non-overlapping. Orthogonal channels each capture a distinct dimension without duplicating one another.

Parkinson range. A volatility measure based on the gap between an hour’s highest and lowest prices.

Persistence baseline. A simple forecast that just repeats the last observed value (“the next value equals the current one”). Models are judged by how far they beat it.

p-value. A number indicating how likely a result could have appeared by chance. Small values (for example below 0.05) suggest the result is real rather than accidental.

Quantile. A cut-point in a range of possible values (the median is the 50% quantile). The model predicts a spread of quantiles to express its uncertainty.

Regime, structural break. A regime is a distinct market state (for example calm versus wartime). A structural break is the point where the market shifts from one regime to another.

Risk premium. The extra compensation markets demand for bearing risk. Here, how much a news item raises uncertainty about oil supply.

Salience. How attention-grabbing or prominent something is, regardless of whether it is good or bad.

Sentiment score. A numerical reading of how positive or negative a text is. In this thesis it specifically means the net effect the news implies for the WTI price.

Structured extraction. Having a model fill in a fixed set of fields (a schema) from free text, instead of returning a single label.

Temporal Fusion Transformer (TFT). A neural-network forecaster for time series designed to be interpretable, so one can read off which inputs and which past hours it relied on.

Tone versus price impact. Tone is how positive or negative the words of an article sound. Price impact is what the news actually implies for the price. The two can disagree: a supply threat sounds alarming (negative tone) yet pushes prices up (bullish).

VAR (vector autoregression). A time-series model of how several variables move together and influence one another over time.

Variable Selection Network (VSN). The part of the Temporal Fusion Transformer that learns how much weight to give each input feature.

VIX. A market index measuring the stock market’s expected volatility, often called the “fear gauge”.

Volatility. How much a price fluctuates. Higher volatility means larger, faster swings.

Volume (trading volume). The number of contracts traded in a period. It is the primary liquidity measure in this thesis.

WTI (West Texas Intermediate). The U.S. benchmark grade of crude oil, and the instrument this thesis studies.

1. Introduction

1.1 Motivation

News moves markets. That media sentiment forecasts not only prices but trading activity is well established: media pessimism predicts downward price pressure and elevated trading volume [Tetlock, 2007]. Yet most empirical work on news and markets operates at daily resolution, aggregating away the intraday window in which information actually propagates into trading. The market response to a scheduled release is concentrated within minutes to hours of its arrival and decays over the rest of the session [Ederington and Lee, 1993]; a daily bar cannot resolve whether liquidity reacts on impact, an hour later, or over the following trading day.

WTI crude oil is an especially informative setting for closing that gap. Its price and trading activity are driven by a rich, well-dated event structure, OPEC policy, geopolitics in the major producing regions, the weekly EIA inventory cycle, and broader macroeconomic releases, each of which arrives at a known time and plausibly moves liquidity at a horizon shorter than a day. This combination of frequent, identifiable news and a deeply liquid market makes WTI well suited to an hourly-resolution study of how news propagates into liquidity, which daily-frequency work has been unable to observe.

1.2 Problem statement

This thesis asks how news propagates into WTI crude oil liquidity at hourly resolution. Three concerns organize the problem. First, the lag structure: when news arrives, does trading activity react in the same hour, or with a delay, and if delayed, over what horizon? Second, directional asymmetry: does negative (bearish) news move liquidity more than positive (bullish) news of comparable magnitude, as a loss-aversion account would predict? Third, the role of context: how much of the liquidity response is attributable to the news signal itself versus the macro-financial state (volatility, the dollar) and the type of event, and can that attribution be read off an interpretable model rather than assumed?

Answering these requires turning unstructured news text into numerical features at hourly resolution and aligning them with a market grid, then modeling the news-liquidity relationship with methods whose assumptions are transparent enough to support interpretation of

the lag and asymmetry structure.

1.3 Research questions

RQ1 (lag structure). At what lag, if any, does news sentiment exert its strongest effect on WTI trading liquidity, measured at hourly resolution?

RQ2 (directional asymmetry). Does the liquidity response differ between bearish and bullish news, and if so, in what sense?

1.4 Contributions

This thesis makes three contributions, developed in Chapters 4 and 5.

(a) An empirical lag-and-asymmetry characterization for WTI at hourly resolution. The news-liquidity response is not contemporaneous: its strongest effect falls at a lag of roughly six to twelve hours, recovered independently by a lag-regression baseline (Section 4.2.3) and by the attention pattern of a Temporal Fusion Transformer (Section 4.3.7). On asymmetry, the two phases estimate different quantities and, read together, sharpen rather than contradict each other: a linear baseline finds a larger marginal coefficient on negative than on positive sentiment (Section 4.2.3), whereas the forecasting model shows no bearish-versus-bullish gap in predicted volume and instead keys on the risk the news implies, volatility, the extracted risk-premium channel, and supply-risk entities, rather than on the sign of the sentiment (Section 4.3.7). The finding is therefore not that bearish news beats bullish news, but that liquidity responds to the salience and risk content of news more than to its direction.

(b) A channel decomposition of LLM-extracted news features, validated by cross-model agreement. Rather than collapse each article onto a single sentiment score, Phase 2 extracts orthogonal supply, demand, and risk-premium channels with a large language model, mirroring the structural decomposition of oil-price shocks [Kilian, 2009]. In the absence of human ground truth, the extraction is validated by inter-model agreement: recalibrating onto the channels raises cross-model correlation on the composite sentiment from 0.39 to 0.88 (Section 4.3.4). The decomposition also resolves an apparent paradox, why supply-risk geopolitical news is price-bullish rather than bearish.

(c) A comparison of title-only and full-body news representations. Sentiment extracted from headlines alone disagrees with sentiment from full article bodies on 41.6% of articles, and the disagreement is systematic: flipped articles shift bearish far more often than bullish (Section 4.2.2). This quantifies a headline bias that studies relying on titles alone silently inherit.

Underlying all three is a methodological template: a simple, interpretable baseline (FinBERT features with lag regression) and a richer, more expressive second phase (LLM channels with a Temporal Fusion Transformer) that corroborates the baseline and enriches its analytical vocabulary, with neither phase discarded (Section 5.3).

1.5 Structure of the thesis

Chapter 2 fixes terminology and situates the thesis against prior work on news and markets, financial-text models, and interpretable forecasting. Chapter 3 details the two-phase methodology: data acquisition and hourly alignment, feature extraction (FinBERT in Phase 1, LLM channel extraction in Phase 2), and the modeling and evaluation setup. Chapter 4 reports the experiments: the headline-bias comparison and lag-regression results of Phase 1, and the Temporal Fusion Transformer results of Phase 2, including feature importance, temporal attention, and the directional-asymmetry test. Chapter 5 interprets the findings against the research questions and states the study’s limitations. Chapter 6 concludes. The appendices provide the extraction prompt and schema, the ablation and hyperparameters, the canonical entity list, and a reproducibility statement.

2. Background

This chapter provides the terminology and prior work needed to follow the methods (Chapter 3) and experiments (Chapter 4): the WTI market and its liquidity measures, the NLP tools used to turn news into features, the Temporal Fusion Transformer, and the strands of literature that motivate the design choices of this thesis.

2.1 Terminology

This section fixes the vocabulary used throughout; each concept is developed at its point of use in Chapters 3 and 4.

WTI crude oil futures and the EIA inventory cycle. West Texas Intermediate (WTI) is the U.S. benchmark grade of crude oil; this thesis studies its front-month futures contract, the most actively traded WTI instrument. WTI is chosen for its deep liquidity, its sensitivity to a well-defined event structure (geopolitics, OPEC policy, macroeconomic releases), and the weekly cadence of the U.S. Energy Information Administration (EIA) Weekly Petroleum Status Report, which publishes national crude inventory levels every Wednesday at 10:30 a.m. Eastern Time and is the most regular scheduled event in the oil calendar.

Liquidity measures. Market liquidity is the ease of trading an asset without moving its price; this thesis quantifies it hourly with three complementary measures. Trading volume, used in log form, captures raw activity. The Amihud illiquidity ratio [Amihud, 2002], the absolute return per unit of volume, captures price impact, how far the price moves for a given amount of trading. The Parkinson range [Parkinson, 1980], the log high-low range within an hour, is a high-frequency estimator of volatility. Together they describe activity, price impact, and volatility.

NLP for finance. Turning news text into numerical features draws on three natural-language-processing ideas. Three-class sentiment classification labels a text as positive, neutral, or negative, the standard output of financial sentiment models. Structured extraction instead asks a model to fill a fixed schema of fields (for example a sentiment score, an event type, and a list of entities), yielding richer, multi-dimensional features. Prompt engineering, and its stricter form of tool-use schema enforcement, is how a large language model is instructed to produce that structured output reliably.

Temporal Fusion Transformer. The Temporal Fusion Transformer (TFT) [Lim et al., 2021] is an attention-based neural architecture for multi-horizon time-series forecasting. Two of its components give this thesis its interpretability: the Variable Selection Network, which assigns each input feature a learned importance weight, and the interpretable multi-head attention layer, which exposes how much the model attends to each past time step when forecasting. These let the model’s reasoning be read off directly rather than treated as a black box.

2.2 Related work

The design decisions of this thesis are grounded in prior work rather than derived in isolation. This section states the anchors; each is picked up at the point of use in Chapters 3 and 4.

News sentiment and market activity. The premise that news sentiment moves trading activity, and not only prices, is established in the finance literature: media pessimism has been shown to predict downward price pressure and elevated trading volume [Tetlock, 2007]. This grounds the use of news sentiment as a predictor of hourly liquidity in Phase 1 (Chapter 4, Section 4.2).

News at intraday resolution. The market response to news is concentrated at high frequency rather than spread evenly across the day: scheduled macroeconomic announcements account for most of the intraday and day-of-week volatility patterns in interest-rate and foreign-exchange futures, with the bulk of the adjustment occurring within minutes of release and volatility remaining elevated for several hours afterward [Ederington and Lee, 1993]. Most news-sentiment studies in finance nonetheless operate at daily resolution, aggregating away exactly the intraday structure this thesis sets out to measure. Working at hourly resolution (Chapter 3) is the central methodological choice that follows.

News in oil markets. The price response of oil to news is muted at coarse resolution: daily energy-price changes show no compelling response to daily U.S. macroeconomic news releases, consistent with energy prices being predetermined with respect to macro aggregates at the daily horizon [Kilian and Vega, 2011]. A null response at daily frequency does not preclude one at finer resolution, and it motivates this thesis’s twin choices of hourly data and a liquidity target (trading volume) rather than price.

Financial-text sentiment models. Domain-adapted transformer models outperform general-purpose sentiment tools on financial language; FinBERT [Araci, 2019] is the de facto baseline for financial sentiment and is adopted as the Phase 1 sentiment extractor (Section 3.3.1). Its coarse three-class output is also the limitation that motivates the richer LLM-based extraction of Phase 2.

Liquidity measurement. Hourly liquidity is quantified with standard market-microstructure measures: trading volume, the Parkinson high-low range [Parkinson, 1980], and the Amihud illiquidity ratio [Amihud, 2002], which captures price impact per unit of traded volume.

Oil-price shock decomposition. Oil-market movements are not a single undifferentiated shock but a composition of separable drivers: the structural decomposition of the real oil price into supply, aggregate-demand, and precautionary (risk) demand components [Kilian, 2009] is the canonical framework. This grounds the Phase 2 channel decomposition (Sections 3.3.4 and 4.3.2), which extracts a `supply_impact`, `demand_impact`, and `risk_premium` from each article rather than collapsing them onto a single sentiment axis, and it clarifies why supply-risk geopolitical news is price-bullish rather than bearish.

Interpretable multi-horizon forecasting. The Temporal Fusion Transformer [Lim et al., 2021] is an attention-based architecture for multi-horizon time-series forecasting that couples predictive accuracy with interpretability through a Variable Selection Network and interpretable multi-head attention. These two mechanisms are what let Phase 2 (Sections 3.6, 4.3.5, and 4.3.7) read feature importance and temporal attention off the model instead of treating it as a black box, and the thesis uses them directly to corroborate the lag structure of RQ1.

LLM-based extraction and cross-model validation. Large language models can extract structured, multi-dimensional features from financial text, and they can stand in for costly human annotation, matching or exceeding crowd workers on text-labelling tasks [Gilardi et al., 2023], and can serve as evaluators of other models’ outputs [Zheng et al., 2023]. Phase 2 uses both ideas: Claude Haiku extracts the structured schema of Section 3.3.2, producing the richer, price-oriented, channel-decomposed features that FinBERT’s three-class tone label cannot, and the inter-model calibration of Section 3.7 (Section 4.3.4) validates those features by cross-model agreement against a second model.

3. Methods

3.1 Data sources and acquisition

This thesis combines three distinct data sources into a unified hourly dataset for WTI crude oil: market data on prices and trading activity, structured macroeconomic indicators, and unstructured news content. Each source has its own native cadence and access pattern, and aligning them onto a common time grid is a central methodological question, addressed in Section 3.2.

3.1.1 Commodity selection

WTI crude oil futures were chosen as the target commodity after preliminary screening of candidates. Sugar futures were initially considered but excluded due to limited price variability and weak responsiveness to news events at hourly resolution, both of which are essential for the research questions of this thesis. WTI, in contrast, has richer event structure driven by geopolitics, OPEC policy, the weekly EIA inventory cycle, and broader macroeconomic forces, all of which produce measurable hourly volume and price reactions.

3.1.2 Market data

Hourly OHLCV data are retrieved via the yfinance Python library [Aroussi, 2024] for three instruments:

- CL=F: WTI crude oil front-month futures contract, the primary target series
- DX-Y.NYB: the U.S. Dollar Index (DXY), used as a macroeconomic covariate
- ^VIX: the CBOE Volatility Index, used as a market-stress covariate

DXY and VIX enter as exogenous controls in the Phase 2 modeling pipeline (Section 4.3), allowing the news signal to be isolated from broader macro-financial dynamics. They are not used in the Phase 1 statistical analyses (Section 4.2), which focus on the direct relationship between news sentiment and trading volume without macro controls.

From the raw OHLCV records, we derive three liquidity and volatility proxies and one return measure at hourly resolution:

- **log_volume**: the natural logarithm of contracts traded per hour, used as the primary liquidity measure
- **price_range**: the hourly Parkinson range [Parkinson, 1980], computed as $\log(\text{high}) - \log(\text{low})$, a high-frequency volatility proxy
- **log_return**: the hourly log return, $\log(\text{close}_t / \text{close}_{t-1})$
- **amihud**: the Amihud illiquidity ratio at hourly resolution, computed as $|\log_return| / \text{volume}$ following Amihud [2002]. The ratio is undefined for hours with zero volume; such rows are excluded from analyses where Amihud is the dependent variable.

The canonical market grid spans 13 May 2024 to 13 May 2026, yielding 11,232 hourly observations across roughly 24 months of trading. The Phase 1 analysis (Section 4.2.1) ran on an earlier market pull that reached further back, approximately March 2024 to March 2026 (11,219 aligned market hours), and those values are preserved with the Phase 1 features. The two windows differ because yfinance serves hourly data only for a rolling window of about the last two years: by the time the market data was re-pulled and the corpus extended through the post-war period for Phase 2 (Section 4.3.1), the earliest hours of 2024 had dropped out of that window, so the Phase 2 grid begins in May 2024. The roughly 13-hour gap between the two counts is immaterial to any result and reflects only this re-pull, not a change in methodology.

3.1.3 EIA inventory data

Weekly U.S. commercial crude oil inventory reports are retrieved from the U.S. Energy Information Administration (EIA) v2 API [U.S. Energy Information Administration, 2024]. We query the WCRSTUS1 series (weekly ending stocks of crude oil and petroleum products, U.S. total), specifying `frequency=weekly`. The full historical series from January 2020 onward is downloaded to allow computation of rolling baselines, even though only the 2024–2026 window is used in modeling.

Each weekly release is converted into two derived features available at hourly resolution after temporal alignment:

- **is_eia_release**: a binary indicator that fires on the trading hour containing or immediately following the official release time
- **eia_surprise**: the weekly inventory change relative to a rolling four-week baseline, capturing the directional surprise component of each release

EIA releases occur on Wednesdays at 10:30 a.m. U.S. Eastern Time and are a regularly scheduled, closely watched event in oil markets, which is why we derive EIA features

from them. In the reported models, however, the EIA cycle enters only through the `is_wednesday` calendar proxy (Section 3.6.2): the `is_eia_release` and `eia_surprise` features were computed and stored, and tested in an additional model run, but the continuous surprise carried negligible feature importance and the release flag added little beyond the calendar proxy, so neither is part of the canonical model.

3.1.4 News data

Unstructured news content is sourced from the GDELT Project [Leetaru and Schrod, 2013], a large-scale machine-monitored news corpus covering broadcast, print, and web sources worldwide. We use the GDELT 2.0 Document API to retrieve article metadata (URL, source, publication time, title) across a set of thematic queries chosen to capture WTI-relevant content.

The query set evolved across the two methodological phases. In Phase 1, the focus was on direct oil-market terms, yielding eight queries:

```
crude oil market
OPEC production cut
oil supply demand
petroleum inventory
Brent crude price
oil inventory report
energy market outlook
crude oil WTI price
```

The Phase 1 scrape produced 51,948 raw article records across the March 2024 to February 2026 window. After deduplication and English-only filtering, 16,326 unique articles remained. Article body content was retrieved by HTTP-fetching each URL and parsing the page. Body retrieval achieved an approximate 80% response rate, with failures concentrated on paywalled domains, Cloudflare-protected sources, and pages requiring JavaScript rendering. An allow-list heuristic (Section 3.4) classified 7,756 of the retrieved bodies as substantive content. The Phase 1 analyses operated on a combined dataset of 13,690 articles: 7,756 with valid bodies feeding title-plus-body sentiment extraction, and 5,934 articles where body retrieval failed but the title alone was retained as a fallback input.

In Phase 2, the query set was expanded to capture geopolitical and demand-side themes that became prominent in the post-war coverage period (Section 4.3.1):

```
Iran sanctions oil
Saudi Arabia oil production
China oil demand
```



```
Russia oil exports
oil supply disruption
```

The expanded scrape, re-run over the full window through May 2026 with both query sets, produced 76,345 raw article records spanning January 2024 to May 2026. After deduplication and English-only filtering, 22,795 unique articles remained, the modeling-ready Phase 2 corpus (Section 4.3.1). Across this corpus, 19,619 articles produced a real body; the remaining 3,176 yielded either a null response, a Cloudflare or paywall block, or an explicit fetch error. The choice to forward articles without bodies into LLM-based filtering rather than retain a title-only fallback path is discussed in Section 3.4.

3.1.5 Persistence

All retrieved data are stored in a single SQLite database (`wti_thesis.db`) with the following primary tables: `articles` (raw GDELT records and scraped bodies), `market_context` (the hourly grid of derived WTI liquidity features together with the DXY and VIX levels), `eia_events` (raw EIA releases), `llm_features` (LLM-extracted news features, Phase 2 only), `article_entities` (canonical entity mentions per article, used to build the entity flags), and `liquidity` (the modeling-ready joined view, populated by the alignment step in Section 3.2). Storing all data in a single relational database, rather than in CSV intermediates as in the original Phase 1 pipeline, eliminated state drift across notebooks and made the canonical alignment computable as a single deterministic join.

3.2 Temporal alignment

The three data sources operate on different clocks. Hourly market data lives on a clean grid of trading hours, with each hourly bar tagged by its closing timestamp. EIA releases are weekly events at fixed times. News articles, however, can be published at arbitrary timestamps including overnight hours, weekends, and holidays when no trading is taking place. To support models that condition liquidity at hour t on news published before t , each article must be assigned to a specific hour in the market grid.

We address this with two design choices, applied at insertion time when articles are written to the database: ceiling assignment to the next trading hour, and forward-assignment of off-hours articles to the next available trading hour.

3.2.1 Ceiling assignment to the next trading hour

For each article with publication timestamp τ , we compute its assigned market hour as

$$\text{datetime_hour}(\tau) = \lceil \tau \rceil_{1h}. \quad (3.1)$$

That is, an article published at 14:23 UTC is assigned to the 15:00 hourly bar, not the 14:00 bar. The motivation is causal: the 14:00 hourly bar covers the period from 14:00 to 15:00, and its volume is partially formed before 14:23. Assigning the article to the 14:00 bar would create a reverse-causal contamination, allowing news to influence a candle whose formation preceded the news publication. The ceiling rule guarantees that the article strictly precedes the start of the trading hour it is matched to.

An earlier version of the pipeline used `round` rather than `ceil`, which produced the same causal leak for articles published in the second half of an hour. The fix was applied early in the project and propagated through all downstream notebooks.

The ceiling rule is conservative: it assigns news to the next full hour even when the article would plausibly have moved markets within minutes of publication. This may underestimate the immediate (intra-hour) reaction. However, it eliminates the alternative bias of attributing pre-publication price movements to post-publication news, which is the more serious threat to causal inference. This conservative direction of the bias, underestimating rather than inflating the immediate reaction, is noted again where the lag results are interpreted (Section 4.2.3).

3.2.2 Forward-assignment of off-hours articles

A substantial portion of the news corpus is published during hours when no trading is taking place: overnight, on weekends, or during holiday closures. Two strategies were considered. The first is to drop such articles entirely. The second is to forward-assign them to the next available trading hour. We adopted the second.

The motivation is empirical. Geopolitical events, supply shocks, and weekend OPEC communiqués are routinely covered during off-hours and are precisely the events that drive the largest market reactions when trading resumes. Dropping them would systematically remove the most informative subset of the corpus. Forward-assignment, by contrast, preserves the information at the cost of introducing a measurement-time lag between publication and the assigned hour.

Concretely, for each article whose ceiling-assigned hour does not appear in the market grid (because the market is closed), we assign the article to the next available `market_context` row in chronological order. This is implemented as a vectorised pandas `merge_asof` join with `direction='forward'`, which performs the lookup efficiently across the full corpus in a single pass.

We retain a diagnostic column, `assignment_gap`, defined as the difference in hours between

the article’s assigned market hour and its original publication timestamp:

$$\text{assignment_gap} = \text{datetime_hour} - \text{publication_timestamp}. \quad (3.2)$$

For articles published during a trading session, this is at most one hour. For weekend or overnight articles, it can extend to many hours or, in the worst case, across a multi-day holiday closure. The gap is preserved in the `liquidity` table and is used downstream both as a control variable and as a basis for stratified analyses that distinguish *contemporaneous* news ($\text{gap} < 2\text{h}$) from *forward-assigned* news ($\text{gap} \geq 2\text{h}$). The exploratory Phase 1 VAR (later abandoned, Section 4.2.4) was restricted to contemporaneous articles aggregated to hourly resolution; the TFT models in Phase 2 use the full corpus and let the model learn to weight contemporaneous and forward-assigned articles differently if the data support such a distinction.

3.2.3 Joined output

The alignment step produces the `liquidity` table, in which each row is one article enriched with the market-context features at its assigned hour. One boundary effect is worth noting: under the forward-assignment rule (Section 3.2.2), articles published before the market grid begins are mapped to the first available trading hour rather than dropped. In the Phase 2 corpus this places roughly 3,600 pre-coverage articles (published before May 2024) onto the first grid hour, which the models therefore see as a single unusually high-count hour at the very start of the series. Excluding pre-coverage articles outright removes this artifact, an adjustment applied in the v2.3 ablation.

The resulting `liquidity` table has 22,795 rows, of which 15,290 are contemporaneous ($\text{gap} < 2\text{h}$) and 7,505 are forward-assigned ($\text{gap} \geq 2\text{h}$). These two subsets are tracked separately in some analyses but combined in the primary modeling.

3.3 News feature extraction approaches

News articles are unstructured text; downstream statistical and deep learning models require numerical features. The transformation from raw article body to model-ready features is the central NLP problem of this thesis. We use two distinct approaches across the two methodological phases of the project: FinBERT-based sentiment classification in Phase 1, and structured LLM extraction with Claude Haiku in Phase 2. The two approaches differ both in their output representation and in their conceptual model of what a “news feature” is.

3.3.1 FinBERT sentiment extraction

In Phase 1, we use FinBERT [Araci, 2019], a BERT variant fine-tuned on financial text for three-class sentiment classification (positive, neutral, negative). FinBERT is among the most widely used sentiment models in financial NLP and provides a natural baseline for comparison.

The model is applied via the HuggingFace Transformers library [Wolf et al., 2020], running on Apple Silicon GPU acceleration via the MPS backend. For each article, the model returns softmax probabilities over the three classes; the predicted label is taken as the argmax of these probabilities, and a scalar **confidence** value is recorded as the maximum class probability:

$$\text{confidence} = \max(P_{\text{positive}}, P_{\text{negative}}, P_{\text{neutral}}). \quad (3.3)$$

The categorical label, all three class probabilities, and the confidence value are retained for downstream use (persisted in the Phase 1 feature table `gdelt_wti_sentiment.csv`).

FinBERT is applied twice per article:

- **Title-only input:** the article title fed alone to the model
- **Title-plus-body input:** the title concatenated with the body, subject to FinBERT’s 512-token input limit

Body content is truncated where necessary, with the title preserved at the start of the sequence so that the most salient signal is never lost to truncation. Articles for which body retrieval failed (Section 3.1.4) fall back to title-only sentiment as the canonical input, with the body-availability status preserved in the database so downstream analyses can distinguish the two input regimes.

For the lag OLS analysis (Sections 3.5.1 and 4.2), we use the FinBERT class probabilities directly rather than the argmax label, so that each article is weighted by the model’s confidence instead of collapsing to a hard vote. The lag OLS uses $P(\text{negative})$ and $P(\text{positive})$ as two separate continuous regressors, which preserves the bearish-versus-bullish asymmetry that RQ2 requires. The earlier discrete mapping $\{\text{positive} : +1, \text{neutral} : 0, \text{negative} : -1\}$, which discards the probability magnitudes, is retained only as a baseline for comparison in Section 4.2.3. The title-only probabilities are reserved for the comparative analysis in Section 4.2.2 (the headline bias finding), where they support both a categorical divergence measure and a continuous one.

3.3.2 Structured LLM extraction with Claude Haiku

In Phase 2, FinBERT is replaced by Claude Haiku 4.5 [Anthropic, 2025a], a smaller-scale member of Anthropic’s Claude model family, used to extract a richer set of structured features from each article. The motivation for moving from FinBERT to an LLM is taken up in detail in Chapter 4 (Section 4.3.2); briefly, the three-class categorical output of FinBERT collapses information that is naturally continuous and multidimensional, and the confidence value (the maximum class probability) carries limited downstream signal.

The Haiku extraction underwent one schema iteration during Phase 2. The initial schema (referred to as Haiku v1) was used to produce features for the first TFT training (Section 4.3.5) on the Phase 1 corpus of articles. It returned, for each article: a continuous `sentiment_score` in $[-1, +1]$ measuring the article’s net directional impact on the WTI price (-1 strongly bearish, $+1$ strongly bullish, assessed from the article’s content rather than its headline tone), a `magnitude` score for event importance, a `price_direction` label, an `event_type` label, a list of `entities`, a `certainty` score, and a `time_horizon` label. The revised schema (referred to as Haiku v2) was introduced in response to an inter-model calibration finding documented in Section 4.3.4. It dropped the `price_direction` field, changed `event_type` from a single label to an array of one to three labels ordered by salience, introduced an explicit `usable` boolean for filtering (Section 3.4), and added three orthogonal economic channel scores (`supply_impact`, `demand_impact`, `risk_premium`) described in Section 3.3.4. The full Haiku v2 prompt and tool schema are reproduced in Appendix A.

3.3.3 Tool-use API and schema enforcement

The Haiku extraction uses Anthropic’s tool-use API [Anthropic, 2025b] with `tool_choice` forcing the model to invoke a single `extract_article_features` tool. This is not a stylistic choice; it is a methodological one. Tool-use enforces the output schema at the API level, including enumerated values for categorical fields, type constraints on numeric fields, and required-field validation. An earlier extraction pass that relied on the LLM to produce valid JSON via prompt instruction alone produced frequent schema violations (e.g., the model returning `slightly_bullish` instead of a numeric score, or an out-of-range `time_horizon` label). Tool-use enforcement validates the numeric ranges and the scalar enumerated fields at the API boundary and removes almost all such failures. A negligible residue survives where the schema is permissive: the free-text `event_type` array received an off-list label on 78 of 10,514 articles (0.74%, `structural`), and one article carries a malformed `time_horizon` string; both are treated as noise downstream. The enforcement therefore eliminates the bulk of the parsing brittleness rather than every last case.

The tool-use design also enables a short-circuit optimisation. When the model determines

that an article is unusable (a paywall block, a cookie notice, or off-topic content), only the `usable` field is required to be set; all other fields are optional. The model exploits this to return roughly 10 output tokens instead of 200 on unusable articles, reducing the cost of unusable-article extraction by roughly 90% at the per-article output level.

3.3.4 Channel decomposition

The Haiku schema includes three continuous scores beyond the composite sentiment value: `supply_impact`, `demand_impact`, and `risk_premium`, each in $[-1, +1]$. These three channels are designed to decompose the economic content of a news article into orthogonal dimensions:

- `supply_impact` measures the article’s implication for the quantity of oil available to the market. Positive values indicate supply increases (sanctions lifted, OPEC adding barrels, new pipeline online); negative values indicate supply decreases (refinery outages, OPEC cuts, infrastructure attacks).
- `demand_impact` measures the article’s implication for oil consumption. Positive values indicate demand strength (Fed easing, China stimulus, growth surprise); negative values indicate demand weakness (recession signals, Fed hawkishness, lockdowns).
- `risk_premium` measures the change in geopolitical or operational risk priced into oil. Positive values indicate escalation (military strikes on oil infrastructure, sanctions imposed); negative values indicate de-escalation (ceasefire, sanctions lifted).

The three channels were not part of the original Phase 2 schema. They were introduced as a response to a documented inter-model calibration failure on the composite `sentiment_score`, in which Haiku and a GPT-family reference model disagreed on the sign of sentiment for high-magnitude geopolitical events. The full reasoning behind the decomposition is documented in Chapter 4 (the calibration failure in Section 4.3.4, and the decomposition with its re-calibration in Section 4.3.6). The methodological role of the channels in this thesis is twofold: they are higher-resolution features for the TFT v2 model (Section 4.3.7), and they serve as a defensible substitute for human-rater agreement when validating LLM-extracted features (Section 4.3.4).

The composite `sentiment_score` is retained alongside the channels rather than replaced by them. Three reasons motivate this. First, it preserves continuity with the Phase 1 headline bias experiment, which depends on a single sentiment value. Second, it allows ablation studies comparing the composite signal against the decomposed channels in the TFT (Section 4.3.7). Third, it provides a hedge against the channels turning out to be individually noisy at production scale.

3.3.5 Magnitude, event type, entities, certainty, and time horizon

The remaining fields in the Haiku schema capture orthogonal dimensions of news content. **magnitude** is a continuous score in $[0, 1]$ measuring how market-moving an event is, with anchors provided in the prompt (0.0 = negligible event, 1.0 = historic; most articles score in the 0.1–0.4 range, with a major OPEC cut anchored at 0.9 and a full-scale geopolitical crisis at 1.0). **event_type** is an array of one to three categorical labels drawn from a fixed set of six values (**geopolitical**, **supply**, **demand**, **macro**, **technical**, **other**), ordered by salience within the article. **entities** is a list of named actors central to the article (countries, organisations, oil companies, key officials). **certainty** is a continuous score in $[0, 1]$ distinguishing rumour, analyst forecast, and confirmed fact. **time_horizon** is a categorical label drawn from **immediate** (hours to a day), **short_term** (days to weeks), and **structural** (months or longer themes such as energy transition).

Each of these fields encodes information that FinBERT’s three-class output cannot represent, and each enters the TFT v2 model either as a continuous input or as a categorical that the model’s Variable Selection Network can weight independently. The raw **entities** lists are normalised in post-processing to a fixed canonical set of 71 actors through an alias mapping (for example “Tehran” and “Iranian” map to **Iran**); at hourly aggregation each hour receives a binary vector of length 71 flagging which canonical entities the dominant article of that hour named, and this vector enters TFT v2 as 71 entity-flag features (the full list is in Appendix D). The design rationale for each field, including the choice to encode **event_type** as a list rather than a single value and the choice to handle entity normalisation in post-processing rather than in the prompt, is discussed in Section 4.3.2.

3.4 Filtering strategies

A substantial fraction of articles retrieved from GDELT is not substantive news. Common categories of non-substantive content include paywall placeholders, cookie consent notices, Cloudflare blocks, scraping error pages, generic stock-pick listicles, and articles whose body content is unrelated to oil markets despite matching one of the query keywords. Filtering these out is essential, since they would otherwise contribute noise to the sentiment signal and bias the downstream models.

We use two distinct filtering strategies across the two phases. Phase 1 used a regex/keyword heuristic on the article body. Phase 2 replaced this with an LLM-judged binary classification of content usability, exposed as the **usable** field in the Haiku extraction. Both approaches are retained as columns in the database (**body_valid** and **usable** respectively), enabling a direct comparison reported in Section 4.3.3.

3.4.1 Regex/keyword heuristic (Phase 1)

The Phase 1 filter, named `body_valid`, is an allow-list applied to the article body text. A body is considered valid if and only if it satisfies all of the following criteria:

- Minimum length of 400 characters (excluding whitespace and HTML residues)
- Presence of at least one term from a domain-specific keyword list covering energy, financial, and geopolitical vocabulary
- Absence of press-release boilerplate (vendor blacklist for known PR-distribution patterns)
- An all-caps ratio below 30%, computed over words of more than two characters in texts containing more than ten words (excluding scraped pages that are predominantly menu links, category headers, or stock-ticker listings)

The filter was developed iteratively over the early project, starting as a blacklist of known bad phrases (Cloudflare error text, cookie notices) and evolving into the allow-list form described above. On the Phase 1 corpus of 16,326 articles with English text and successful body retrieval, the filter classified 7,756 articles (47.5%) as having valid bodies. The remainder were either retained with title-only sentiment as a fallback or excluded from the Phase 1 analyses depending on the specific notebook (Section 4.2.1 details which analyses used which subset).

The regex heuristic is fast, reproducible, and transparent: any analyst can read the rules and predict their behaviour on a given article. It is also limited in ways that became evident during inspection. Articles passing the filter sometimes contained substantive boilerplate (long disclaimers, repeated company descriptions) that nevertheless mentioned energy keywords. Articles failing the filter sometimes contained legitimate analyst commentary whose body length fell just below the 400-character threshold, or whose vocabulary did not match the energy keyword list closely enough.

These observations motivated the migration to a more capable filter in Phase 2.

3.4.2 LLM-judged usable flag (Phase 2)

The Phase 2 filter, exposed as the `usable` boolean in the Haiku schema, asks the LLM itself to judge whether the article body is substantive WTI-relevant content. The prompt instructs the model to return `usable=false` for paywall placeholders, Cloudflare blocks, cookie notices, or articles that are clearly not about oil markets, macro, or geopolitics relevant to oil. When `usable=false`, the model is permitted (and instructed) to omit all other fields, exploiting the short-circuit optimisation discussed in Section 3.3.3.

The migration from `body_valid` to `usable` is motivated by three considerations. First, the LLM has access to semantic content that a regex cannot match: it can distinguish a substantive analyst piece from a press-release-shaped article that happens to contain energy keywords. Second, the LLM’s judgement is calibrated against the same content model used for the rest of the feature extraction, so the filter and the features are coherent (an article flagged as usable will produce features that are themselves usable). Third, the marginal cost of sending more articles to the LLM is small under batch processing with prompt caching, so a broader filter at the input does not impose a meaningful resource constraint.

The `usable` flag becomes the primary content filter in Phase 2. A stricter variant, `usable_strict`, additionally requires at least one of the three economic channels (`supply_impact`, `demand_impact`, `risk_premium`) to be non-zero, which removes the roughly 1,161 articles the model marked topical yet scored as carrying no material market impact; `usable_strict=1` is the canonical filter for TFT v2 training (Section 4.3.3), while `usable=1` is used for the filter comparison. The `body_valid` column is retained in the database for the comparative analysis reported in Section 4.3.3 but is no longer applied as a filter in any modelling step.

3.4.3 Comparative analysis

Both filter columns coexist in the database for the Phase 2 corpus, enabling a direct disagreement analysis. Of 19,619 Phase 2 articles processed by the LLM, 13,550 are flagged `body_valid=1` and 11,675 are flagged `usable=1`. The two filters disagree on a non-trivial proportion of articles in both directions: regex-rejected articles that the LLM accepted as substantive (the regex’s false positives), and regex-accepted articles that the LLM rejected as boilerplate (the regex’s false negatives). The full agreement analysis, including Cohen’s kappa and worked examples of disagreements, is reported in Section 4.3.3.

3.5 Statistical models

The empirical analysis of news impact on WTI liquidity rests on lag-structured ordinary least squares (OLS) regressions, described in Section 3.5.1, which provide the canonical statistical evidence for the research questions on lag structure (RQ1) and bearish-versus-bullish asymmetry (RQ2). An exploratory vector autoregression was also fitted during development but abandoned: the sparse, event-driven sentiment signal (more than half of all hours carry no contemporaneous news) left it unable to identify significant dynamics. It is not developed as a method here; its outcome, and the move to the deep-learning model of Section 3.6 that it motivated, are noted in Section 4.2.4.

3.5.1 Lag OLS specification

The lag OLS analysis tests whether news sentiment published at hour t predicts trading volume at subsequent hours $t + k$, where k ranges over a fixed set of lag horizons. The specification at lag k is

$$\log_volume_{t+k} = \beta_0 + \beta_1 P(\text{negative})_t + \beta_2 P(\text{positive})_t + \varepsilon_t, \quad (3.4)$$

where $P(\text{negative})_t$ and $P(\text{positive})_t$ are the FinBERT title-plus-body class probabilities for the article published at hour t . The neutral probability is the omitted reference: the three class probabilities sum to one, so including all three would be perfectly collinear. The coefficients β_1 and β_2 therefore measure the expected difference in `log_volume` at hour $t + k$ for a maximally confident bearish or bullish article (class probability equal to one), respectively, relative to a neutral one. Using the two class probabilities rather than a single signed sentiment regressor is essential to the asymmetry question. A signed specification would force the model to assume that the bearish coefficient is the negative of the bullish coefficient (a symmetric response); the two-probability specification allows the two coefficients to vary independently, which is precisely what RQ2 requires. This continuous encoding is the primary Phase 1 specification. It generalises the earlier two-dummy encoding (which shares the same two-coefficient structure but with hard 0/1 indicators) by weighting each article by FinBERT’s confidence, so that a 0.95-bearish article contributes more than a 0.51-bearish one; the dummy version is retained as a baseline in Section 4.2.3.

A separate regression is estimated at each lag value $k \in \{1, 2, 3, 4, 6, 8, 12\}$. The lag set is dense at short horizons, where information propagation is expected to be detectable, and sparser at longer horizons, where the effect is expected to attenuate. At each lag, we report the two coefficients, their standard errors, p-values, and the regression R^2 . The R^2 is expected to be small in absolute terms, since news sentiment is one input among many that influence hourly trading volume; the modelling goal is to detect a structured pattern of coefficients across lags rather than to maximise explanatory power. A statistically significant coefficient at the conventional $p < 0.05$ threshold, repeated across multiple adjacent lags with a coherent magnitude pattern, is the criterion for claiming a detectable lag effect.

The contemporaneous specification ($k = 0$) is reported separately as a baseline. At hour t , the article has just been published, so a detectable effect requires the news to act on the market within the same one-hour bar. This turns out to be a small but statistically significant effect (Section 4.2.3): rather than a null result, it establishes that sentiment carries detectable signal from the moment of publication and motivates tracing how that signal evolves across the subsequent lags. Section 4.2.3 reports the contemporaneous result

together with the lag regressions.

All regressions are estimated by ordinary least squares with the default standard error formula. Each observation is a single article. The Phase 1 corpus of 13,690 articles produces between roughly 9,900 and 11,600 per-lag regression rows after dropping articles whose target hour $t+k$ falls outside the coverage window (coverage falls as the lag lengthens). The regression treats these as a cross-section of articles rather than as a sequential time series, so heteroskedasticity- and autocorrelation-consistent (HAC) standard error corrections are not applied. The assumption may slightly underestimate true uncertainty if articles published close in time share unobserved common shocks, but the structured pattern of coefficient significance across multiple lags is robust to this concern: even with moderately wider confidence intervals, the pattern of bearish coefficients peaking at lag +6 hours with bullish coefficients showing the same pattern at lower magnitude would survive.

3.6 Temporal Fusion Transformer

The Temporal Fusion Transformer (TFT) [Lim et al., 2021], introduced by Lim, Arik, Loeff, and Pfister, is a deep-learning architecture for multi-horizon time-series forecasting with mixed continuous and categorical inputs. We adopt the TFT as the primary deep-learning model for the Phase 2 work, training two successive instances (TFT v1 and TFT v2) that share the same architectural skeleton but differ in their feature configuration and training data. This section describes the architecture itself; the feature configurations for each instance are documented in Sections 4.3.5 and 4.3.7.

3.6.1 Architecture overview

The TFT combines four architectural components into an end-to-end model:

- A **Variable Selection Network (VSN)** at each input time step, which assigns a learned scalar weight to each input feature, allowing the model to suppress uninformative features automatically
- A pair of **LSTM encoder–decoder layers** that process the history (encoder) and the forecast horizon (decoder) at fixed resolution
- An **interpretable multi-head self-attention** layer over the encoded sequence, exposing which historical time steps the model attends to when producing each forecast
- A **quantile output head** that produces multiple quantile predictions in a single forward pass, enabling prediction intervals without separate models

Three of these components each contribute a distinct interpretability benefit. The Variable

Selection Network produces per-feature importance scores that are directly comparable across features, addressing the question of which inputs the model actually uses. The attention layer produces per-time-step weights that expose the temporal structure of the model’s reasoning, which is directly relevant to RQ1 on the lag structure of news impact. The quantile output enables uncertainty quantification without separate model training.

3.6.2 Input typing

The TFT distinguishes four categories of inputs:

- **Time-varying known reals:** features whose values are known at every time step in both the history and the forecast horizon. In our setup these are calendar features (`hour`, `day_of_week`, `month`) and binary regime indicators (`is_us_session`, `is_wednesday` as a proxy for the EIA release day).
- **Time-varying unknown reals:** features whose values are known only over the history and must be predicted for the forecast horizon. These include the target itself (`log_volume`), the market-context features (`price_range`, `log_return`, `amihud`, `dxy`, `vix`), the continuous Haiku features (the composite `sentiment_score`, the three channels, `magnitude`, `certainty`), and, in TFT v2, the 71 binary entity flags (see Sections 4.3.5 and 4.3.7 for the per-instance feature lists).
- **Time-varying unknown categoricals:** categorical news features known only over the history. In TFT v2 these are the dominant article’s `event_type` and `time_horizon`, passed as learned embeddings rather than integer codes; encoding them as proper categoricals rather than integers is a design axis whose effect is isolated in the v2 ablation (Section 4.3.7.4).
- **Static categoricals:** features that are fixed for the duration of a time series. In our single-asset configuration this reduces to a single asset identifier (`asset = 'WTI'`).

The Variable Selection Network learns separate gating weights for each of these categories, so a feature that is informative in the history (as an unknown real) but uninformative as a calendar input is correctly weighted across both contexts.

3.6.3 Forecast configuration

Both TFT instances condition each forecast on a fixed 48-hour encoder window, providing two full daily cycles of context. This encoder length is intentional: the lag OLS analysis in Phase 1 (Section 4.2.3) identified a news-impact peak at lag +6 hours, and a 48-hour window comfortably covers both this peak and the second-day “memory” effects observed in earlier TFT runs.

The forecast target and horizon differ between the two instances. TFT v1 (Section 4.3.5) predicts a single target, `log_volume`, one hour ahead. TFT v2 (Section 4.3.7) predicts three targets jointly, `log_volume`, `amihud`, and `price_range`, at four horizons (1, 3, 6, and 12 hours), so its prediction length is 12 hours and it emits a full trajectory rather than a single step; the +6 hour horizon is chosen to coincide with the lag OLS peak.

Both instances are trained with a quantile loss, which produces point predictions at multiple quantiles in a single forward pass and is robust to target outliers without distributional assumptions. TFT v1 uses a single `QuantileLoss` on its one target; TFT v2 uses a `MultiLoss` that sums a `QuantileLoss` over each of the three targets. The median quantile serves as the point forecast throughout.

3.6.4 Architectural hyperparameters

The TFT architecture exposes a small set of hyperparameters governing model capacity. The core width parameters are held constant across both instances, so that any difference between them comes from the features and the forecast configuration rather than from raw capacity:

- `hidden_size = 32`: the internal representation dimension
- `attention_head_size = 4`: the number of parallel attention heads
- `hidden_continuous_size = 16`: the embedding dimension for continuous variable processing
- `learning_rate = 1e-3`: the initial Adam optimizer learning rate, with on-plateau reduction (patience 3)

Two things differ between the instances. The `dropout` rate was raised from 0.1 in TFT v1 to 0.15 in TFT v2, a modest increase in regularization for the larger v2 input space. And although the core width is unchanged, the trainable-parameter count grows from approximately 113,000 in v1 to 298,329 in v2, because v2 adds the 71 entity-flag inputs, the two additional targets, and the multi-horizon output head. Holding the width fixed while letting the feature set and forecast configuration expand is deliberate: it keeps the two instances comparable on architectural capacity while isolating the effect of the Phase 2 feature investments.

3.6.5 Training procedure

Training is performed in PyTorch Lightning [Falcon and The PyTorch Lightning team, 2024] via the `pytorch-forecasting` library [Beitner, 2024]. Both instances share the same optimizer and regularization scaffolding: Adam with the learning rate above, gradient clip-

ping at 0.1 to stabilize updates over long sequences, early stopping on validation loss with a minimum improvement of $1\text{e-}4$, and a Google Colab T4 GPU. Fixed-length sliding-window samples are built by the `pytorch-forecasting TimeSeriesDataSet` wrapper, which encodes categorical features consistently, constructs the input tensors, and applies group normalization; the held-out sets use the same wrapper with `stop_randomization=True` for deterministic evaluation.

Three training settings differ between the instances, reflecting their different scale:

- **Temporal split.** TFT v1 uses an 80/20 split, the earlier 80% of hours for training and the most recent 20% for validation, with no held-out test set. TFT v2 uses a 60/20/20 split into train, validation, and test, with 48-hour buffers between partitions to keep encoder windows from leaking across boundaries; the test set is what enables the pre-war versus war-regime evaluation of Section 4.3.7.3.
- **Early stopping and epoch cap.** TFT v1 uses a patience of 5 epochs and a maximum of 30; TFT v2 raises the patience to 10 and the cap to 75. In practice early stopping halts both well before the cap.
- **Target normalization.** TFT v1 normalizes its single target with a `GroupNormalizer`. TFT v2, being multi-target, wraps one `GroupNormalizer` per target in a `MultiNormalizer`, so each of the three targets is scaled independently before the per-target losses are combined.

For reproducibility, the reported TFT v2 run also fixes the random seed (42) and uses deterministic algorithms.

3.6.6 Evaluation and interpretation outputs

For each trained TFT we produce a set of evaluation and interpretability outputs, all reported in Chapter 4.

Performance against a persistence baseline. The headline performance measure is not the raw validation loss. The two instances are trained with different loss functions on different splits and horizons, so their loss values are not directly comparable (Section 4.3.8). Instead, forecasts are scored by mean absolute error (MAE) against a persistence baseline, the naive forecast that carries the current value forward to the target horizon ($\hat{y}_{t+k} = y_t$), a standard reference for financial time series. The reported quantity is the percentage reduction in MAE over persistence, which normalizes for the differing difficulty of each target and horizon. For TFT v2 these metrics are computed for every target (`log_volume`, `amihud`, `price_range`) and horizon (1, 3, 6, 12 hours) on four slices: the validation set, the full test set, and the pre-war and war portions of the test set, split at the Iran war onset (Section 4.3.1). The pre-war and war slices measure how the pre-war-trained model

generalizes across the regime boundary.

Interpretability. Two interpretability outputs are read directly off each model, both averaged over the evaluation set: the Variable Selection Network feature-importance ranking, showing which inputs the model relied on most, and the multi-head attention weights, giving a per-lag map of which historical hours the model attended to.

Research-question diagnostics. The attention output is of particular interest for RQ1: if the TFT attention peaks at a lag consistent with the lag OLS result of Phase 1, the two analyses corroborate one another from very different starting points, the OLS detecting a structured lag effect from a sequence of single-lag linear specifications and the TFT learning a single nonlinear function over the full history while exposing the same structure through attention. For RQ2 we run a directional asymmetry test on the forecasts: hours are split by the sign of their sentiment score into bearish and bullish groups, and a Welch t-test compares the mean predicted `log_volume` of the two groups at each horizon and slice. This is deliberately a different estimand from the lag OLS coefficient comparison, since it contrasts the average level of predicted volume across two groups of hours rather than the marginal sensitivity of volume to sentiment, and Section 4.3.7.6 reads the two together with that distinction in mind.

3.7 Inter-model calibration methodology

The Haiku-extracted feature set described in Section 3.3 is the substrate of all Phase 2 modeling. Before relying on it for downstream analysis, we need an external check that the extraction is producing sensible outputs and that the LLM is not systematically biased on the cases that matter most. This section describes the methodology for that external validation. The specific calibration results, and the methodological refinement that the calibration motivated, are reported in Chapter 4 (Sections 4.3.4 and 4.3.5).

3.7.1 The validation problem and our approach

The standard approach to validating an NLP extraction is to compare model output against human-annotated ground truth on a labelled subset of the data. For this thesis, that approach is not feasible. Human annotation of WTI-relevant news articles requires domain expertise in oil markets and trading that the author does not possess, and a single non-expert annotator working at scale would introduce noise of unknown structure into the validation set rather than reduce it. Recruiting a third-party expert at scale was outside the resource scope of the project.

We adopt instead an inter-model calibration approach, building on evidence that large language models are competent annotators in place of human labels [Gilardi et al., 2023]

and can act as judges of other models’ output [Zheng et al., 2023]: the same articles, with the same extraction prompt and tool schema, are scored by Claude Haiku 4.5 and by a GPT-family reference model (OpenAI’s GPT-5.5, via the ChatGPT interface). Disagreement between two models from different developer families on the same input is informative in a different way than human disagreement, but is informative nonetheless. The argument for treating it as a useful validation signal is the following.

Each model encodes its own implicit notion of news interpretation, shaped by its training data, its alignment procedure, and the family-level inductive biases of its developer. The two models share little in their training data composition or alignment style. When they agree on an extraction, that agreement reflects content that is robust to two independently-trained interpretations of financial news. When they disagree, the disagreement points to either genuine ambiguity in the article or to a systematic bias in one of the models.

This approach is acknowledged as methodologically weaker than expert-annotated ground truth: two models agreeing on a wrong reading would not be detected by the comparison. The limitation is reported as part of the discussion in Chapter 5. It is, however, methodologically stronger than no external validation, and the cross-family choice maximises the diversity of the comparison within the constraint that both annotators are LLM-based.

3.7.2 Sample design

The calibration sample is constructed by stratified random sampling from the full Phase 2 corpus. The strata reflect the regex-based body validity classification used in Phase 1, with three bucket categories:

- **valid**: articles passing the regex `body_valid=1` filter (Section 3.4.1)
- **invalid_substantial**: articles failing `body_valid=1` but whose body field still contains substantial text content (length above a minimum threshold)
- **invalid_weak**: articles failing `body_valid=1` with bodies short enough to be likely scraping artefacts

From each of these three strata, ten articles are sampled uniformly at random, producing a total calibration sample of thirty articles. The stratification ensures that the calibration covers the full range of content the LLM might encounter in production, including the marginal cases where the regex filter and the LLM are most likely to disagree.

3.7.3 Extraction procedure

The same system prompt and tool schema (Appendix A) are submitted to both models for each of the thirty articles. For Claude Haiku, this is the production extraction code

described in Section 3.3.2. For the GPT reference, the prompt and schema are submitted via the ChatGPT interface with manual JSON output extraction.

The two output sets are then aligned by article identifier and compared field by field. The comparison computes agreement metrics symmetrically: no metric depends on which model is treated as the reference.

3.7.4 Agreement metrics

For each field in the schema, we compute one or more of the following agreement measures, chosen to match the field’s type:

- For binary fields (`usable`): agreement rate as a count out of thirty, and Cohen’s kappa to correct for chance agreement
- For continuous fields (`sentiment_score`, `supply_impact`, `demand_impact`, `risk_premium`, `magnitude`, `certainty`): Pearson correlation across all pairs, mean absolute difference (MAD), and the count of **sign disagreements**, defined as cases where the two models’ scores have opposite signs (one positive, one negative). Near-zero scores within a small range around zero are treated as agreement on neutrality and do not contribute to the sign-disagreement count.
- For categorical fields (`event_type`, `time_horizon`): exact-match agreement rate, and for the array-valued `event_type`, the proportion of cases where the top-ranked category from one model appears anywhere in the other model’s list

The sign-disagreement count for continuous fields is the most methodologically important measure. A small mean absolute difference can mask cases where the two models disagree on the qualitative direction of an effect (one rates an event bullish, the other bearish), which is precisely the kind of disagreement that would propagate as noise into downstream models. The sign-disagreement count surfaces these cases explicitly.

3.7.5 Use of the calibration outcome

The calibration sample is small ($n = 30$) by deliberate design: a larger sample would have proportionally larger annotation cost on the GPT side, and the goal is to detect qualitative patterns of agreement and disagreement rather than to produce a precise estimate of population-level agreement rates. The thirty articles are enough to surface systematic disagreements, identify their character, and motivate either acceptance of the schema as-is or schema revision.

In the event the calibration reveals significant disagreement on any field, the response is to investigate the source of the disagreement and revise the extraction prompt or schema

accordingly, then re-run the calibration on the same thirty-article sample with the revised setup. This iterative loop is documented in Chapter 4 (Section 4.3.4 reports the first calibration result, and Section 4.3.6 reports the schema revision and re-runs the calibration with the revised schema). The use of the same thirty articles for both the initial and the revised calibration is intentional: it isolates the effect of the schema change from the effect of sample variation.

4. Experiments

4.1 Overview

This chapter reports the empirical results of both methodological phases of the thesis. Phase 1 (Section 4.2) presents the regression-based analysis of the pre-war corpus, including the headline bias experiment, the contemporaneous and lag OLS regressions that establish the initial evidence for RQ1 and RQ2, and the VAR model. Phase 2 (Section 4.3) presents the deep-learning analysis of the expanded corpus, including the LLM feature extraction pipeline, the Temporal Fusion Transformer models (v1 and v2), and the ablation study that guided the v2 design. The chapter closes with a methodological comparison of the two phases (Section 4.3.8), articulating what Phase 2 changed and why direct numerical comparison between the two phases is not appropriate.

4.2 Phase 1: initial pipeline and baseline modeling

Phase 1 establishes the empirical baseline for the research questions of this thesis, building on prior evidence that news sentiment predicts trading activity and volume, not only prices [Tetlock, 2007]. It uses the FinBERT-extracted news sentiment described in Section 3.3.1 together with the regex-based body validity filter of Section 3.4.1, applied to a corpus of WTI-relevant news articles published between March 2024 and February 2026. Three modeling techniques are applied in turn: linear regression (contemporaneous and lagged), vector autoregression, and a headline-bias comparative experiment that motivates the broader use of full article bodies in downstream feature extraction. By the end of Phase 1, the canonical statistical evidence for RQ1 (lag structure) and RQ2 (bearish-versus-bullish asymmetry) is in hand, and the limitations of the FinBERT-plus-classical-models pipeline are visible enough to motivate the methodological refinements that follow in Phase 2.

4.2.1 Phase 1 data and feature setup

The Phase 1 modeling dataset combines **13,690 articles**: 7,756 with substantive bodies (regex filter `body_valid=1`, Section 3.4.1) feeding title-plus-body FinBERT sentiment, and 5,934 where body retrieval failed but the title was retained as a fallback. The corpus derives from 51,948 raw GDELT records across eight oil-market queries, reduced to 16,326 unique English articles after deduplication (Table 4.1). Each article is scored

by FinBERT [Araci, 2019] (Section 3.3.1); the Phase 1 regressions use the title-plus-body class probabilities as the continuous sentiment feature, with the discrete $\{+1, 0, -1\}$ label kept as a baseline. The sentiment distribution is **44.9% bearish, 30.4% bullish, 24.7% neutral**; the bearish skew reflects the headline-tone tendency documented in Section 4.2.2, not a population-level claim.

Articles are aligned to trading hours by the ceiling rule (Section 3.2.1): 82.7% fall within two hours of a trading hour (contemporaneous), the remainder are forward-assigned. The market backbone is 11,219 hourly WTI records with the liquidity targets (`log_volume`, `price_range`, `amihud` [Amihud, 2002]) defined in Section 3.1; DXY, VIX, and the EIA covariates enter only in Phase 2. Table 4.1 summarises the construction.

Table 4.1: Phase 1 dataset construction.

Stage	Count
Raw GDELT articles	51,948
After deduplication + English filter	16,326
With successful body retrieval	~13,000 (80% of 16,326)
With <code>body_valid=1</code> (regex-passed body)	7,756
Title-only fallback (body failed but title retained)	5,934
Phase 1 modeling dataset	13,690
Aligned market hours (raw, before VAR cleaning)	11,219
Articles aligned within 2h gap (contemporaneous)	11,316
Articles aligned with gap ≥ 2 h (forward-assigned)	2,374
Article date range	2024-03-11 to 2026-03-01

This dataset is the substrate for the three Phase 1 experiments that follow.

4.2.2 Headline bias experiment

Motivation. The Phase 1 pipeline mixes two FinBERT input regimes, title-only (fallback) and title-plus-body, in the same 13,690-article dataset. Before treating the resulting sentiment as one signal, we test whether the two regimes disagree, a question also of independent interest: headlines are written to attract attention, and the body can qualify or even reverse the implication a reader draws from the title alone.

Method and categorical result. For the 7,755 articles with both a title and a substantive body, FinBERT was applied twice (title-only, then title-plus-body), so every difference between the two scores is attributable to the input content, not the model (Section 3.3.1). On the argmax labels the two series **disagree on 41.6% of articles** ($\chi^2 = 2050$, $p < 0.001$), and the disagreement is directionally structured: positive titles keep a positive body in only 59.2% of cases (40.8% shift toward neutral or negative), neutral titles resolve to negative more often than positive (39.6% vs 30.0%), while negative titles stay negative in 78.2%.

Continuous result. The label measure ignores how far sentiment moves and misses shifts inside same-label articles. Defining a signed score $P(\text{positive}) - P(\text{negative})$ per input and the divergence magnitude $|\text{signed_full} - \text{signed_title}|$: label flips carry a mean magnitude of **0.96** (near-full reversals), same-label articles still shift by **0.17** on average (31.6% by more than 0.20), and the mean signed shift from title to body is -0.09 , with 57.6% of articles becoming more bearish once the body is read.

Worked example. The quantities are easiest to read on concrete articles. The signed score is $P(\text{positive}) - P(\text{negative})$, computed once on the title and once on the title plus body; the magnitude is their absolute distance (how far sentiment moved), and the signed shift is $\text{signed_full} - \text{signed_title}$ (which direction it moved).

Table 4.2: Worked examples of title-versus-body sentiment divergence.

Article (title)	signed_title	signed_full	magnitude	signed shift	Label transition
“Oil price soars above \$80”	+0.49	−0.63	1.12	−1.12	positive → negative (flip)
“U.S. Oil Is Stealing Market Share from OPEC+”	−0.32	−0.77	0.45	−0.45	negative → negative
“Oil prices continue to drop in world markets”	−0.96	−0.96	0.00	0.00	negative → negative

The first article flips label: a headline FinBERT reads as bullish (“soars above \$80”) turns bearish once the body is scored, a near-full reversal (magnitude 1.12). The second keeps its negative label yet still moves 0.45 toward more bearish, exactly the within-bucket movement the 41.6% flip rate does not count. The third is stable, title and body agree. In all three the body is at least as bearish as the title, the one-directional pattern behind the -0.09 mean signed shift.

Divergence and label-flip totals. The aggregate quantities across the 7,755 title-plus-body articles are collected in Table 4.3.

Implications. Two findings carry forward. First, the choice of input regime matters: a pipeline that treats title-only and title-plus-body interchangeably averages over a 41.6% disagreement rate. Second, and against the common intuition that attention-grabbing headlines overstate negativity, **titles lean more bullish than the articles they contain:** reading the body moves FinBERT’s sentiment more negative on average, and only the signed continuous score fixes this direction, which the labels alone could not establish. The 5,934 title-only fallback articles therefore carry a sentiment that would have shifted in roughly 40% of cases had a body been available; because that shift is one-directional across all lags, it attenuates rather than inflates the OLS effect reported below. More broadly, this is the first concrete evidence that the choice of news representation matters, and it

Table 4.3: Title-versus-body divergence totals (7,755 articles).

Quantity	Value
Label agreement (title and body same label)	4,527 (58.4%)
Label flip (disagreement)	3,228 (41.6%)
Flip rate, positive titles	40.8%
Flip rate, neutral titles	69.5%
Flip rate, negative titles	21.8%
Mean divergence magnitude, all articles	0.50
Mean divergence magnitude, label flips	0.96
Mean divergence magnitude, same-label	0.17
Same-label articles with magnitude above 0.20	31.6%
Mean signed shift, title to body	−0.09
Articles more bearish once the body is read	57.6%

motivates Phase 2’s move to structured extraction from full article bodies (Section 3.3.2).

4.2.3 Lag OLS: RQ1 and RQ2

The lag OLS regressions are the canonical Phase 1 evidence for RQ1 (lag structure of news impact on liquidity) and RQ2 (bearish-versus-bullish asymmetry), and the result the Phase 2 methodological choices ultimately aim to corroborate.

Specification. At horizon k ,

$$\log_volume_{t+k} = \beta_0 + \beta_1 P(\text{negative})_t + \beta_2 P(\text{positive})_t + \varepsilon_t, \quad (4.1)$$

where $P(\text{negative})_t$ and $P(\text{positive})_t$ are the title-plus-body FinBERT class probabilities for the article published at hour t , with the neutral probability as the omitted reference (Section 3.5.1); β_1 and β_2 measure the `log_volume` response to a maximally confident bearish or bullish article relative to a neutral one. Each article is one observation, and a separate regression is estimated at each $k \in \{0, 1, 2, 3, 4, 6, 8, 12\}$ ($k = 0$ is the contemporaneous case), on 13,690 rows at $k = 0$ falling to roughly 9,900 at the longest lag. The continuous encoding is primary; a discrete-dummy baseline is retained for comparison and, at $k = 0$, gives smaller coefficients ($\beta_{\text{bearish}} = 0.133$, $\beta_{\text{bullish}} = 0.103$) than the continuous 0.186 / 0.166, confirming that weighting each article by FinBERT’s confidence sharpens the signal.

Result. Table 4.4 reports the regression at every horizon. Each row is one regression at lag k (in hours; $k = 0$ is the contemporaneous case). The columns are: $\beta_{P(\text{neg})}$ and $\beta_{P(\text{pos})}$, the fitted coefficients on the bearish and bullish class probabilities, that is, the expected change in `log_volume` for a maximally confident bearish or bullish article relative to a neutral one; the two p columns, their two-sided p -values (a coefficient is significant at the

Table 4.4: Lag OLS coefficients across horizons.

Lag k (h)	$\beta_{P(\text{neg})}$	p	$\beta_{P(\text{pos})}$	p	R^2	n
0 (contemp.)	0.186	0.000	0.166	0.000	0.0015	13,690
1	0.262	0.000	0.200	0.000	0.0028	11,598
2	0.105	0.068	0.114	0.084	0.0003	11,512
3	0.117	0.047	0.170	0.013	0.0006	11,360
4	0.267	0.000	0.237	0.001	0.0017	11,065
6	0.342	0.000	0.291	0.000	0.0027	10,679
8	0.056	0.409	0.101	0.192	0.0002	10,440
12	0.001	0.984	-0.171	0.029	0.0008	9,917

usual threshold when $p < 0.05$); R^2 , the share of `log_volume` variance that the two-feature regression explains at that lag; and n , the number of articles entering that regression, which falls as k grows because targets at hour $t + k$ beyond the coverage window are dropped.

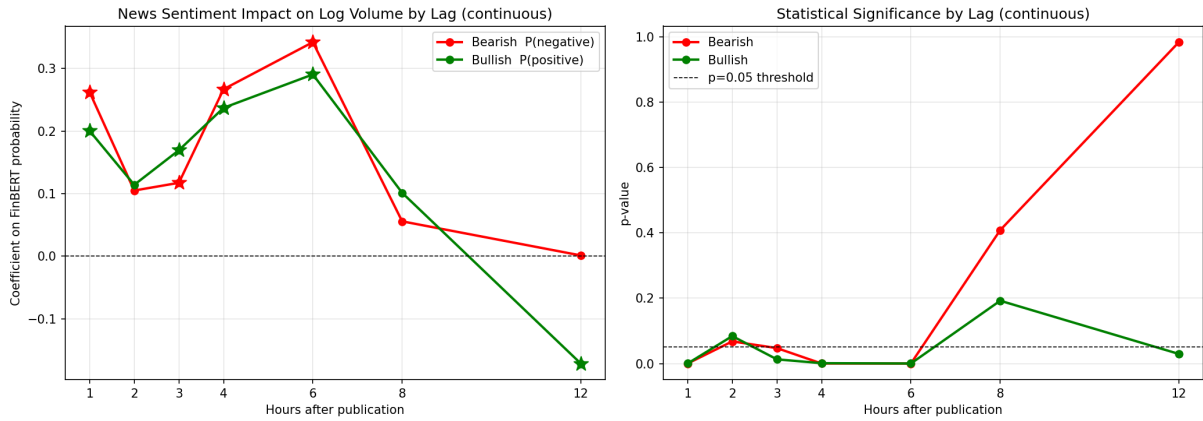


Figure 4.1: Lag OLS coefficients and p-values across horizons. Left panel: the bearish $P(\text{negative})$ and bullish $P(\text{positive})$ coefficients at lags 1, 2, 3, 4, 6, 8, and 12 hours. Stars mark lags significant at the conventional $p < 0.05$ threshold. Right panel: the p-values for the same coefficients, with the $p = 0.05$ reference line. The peak at lag +6h is the canonical finding for RQ1; the ordering of bearish above bullish at the dominant lags is the canonical finding for RQ2.

RQ1 (lag structure). The impact is already detectable at publication ($k = 0$), grows over the first hours, peaks sharply at +6h ($\beta_{P(\text{neg})} = 0.342$, $\beta_{P(\text{pos})} = 0.291$, both $p < 0.001$); the +6h bearish coefficient implies roughly 41% more volume than a neutral hour for a fully confident bearish article, $\exp(0.342) - 1 \approx 0.41$), and decays to insignificance by +8h. The pattern is structured rather than a smooth exponential decay, with secondary maxima at +1h and +4h, suggesting discrete information-incorporation events at characteristic horizons.

RQ2 (asymmetry). Bearish articles produce a stronger response than bullish ones at the dominant lags (0, 1, 4, and the +6h peak, where bearish exceeds bullish by approximately 18%). The one exception is the weak lag 3, where both coefficients are small and bullish is

marginally higher; it does not overturn the pattern at the horizons that carry the effect. Because the two probabilities are estimated as independent regressors (Section 3.5.1), the bearish-over-bullish ordering is not an artifact of an imposed model symmetry.

On the small R^2 and robustness. R^2 stays below 0.003 throughout, which is expected: hourly volume has many drivers and sentiment is one signal, so R^2 measures sentiment’s share of *total* volume variance, not of *news-driven* volume; the substantive finding is the structured coefficient pattern, not the explained variance. Two robustness points. The regression is cross-sectional (one article per row, with no strong serial correlation), so HAC corrections are not applied (Section 3.5.1). And the ceiling alignment guarantees each article precedes its matched hour, which if anything underestimates the immediate reaction, a conservative bias for the peak-lag finding. Both RQ1 and RQ2 are revisited independently by the TFT v2 attention pattern in Phase 2 (Section 4.3.7).

4.2.4 Phase 1 summary

Phase 1 delivers the canonical statistical evidence for both research questions and exposes the limitations that Phase 2 addresses.

Evidence. RQ1 is answered by the lag OLS peak at +6h, and RQ2 by the **bearish** > **bullish** ordering at the dominant lags (Section 4.2.3); both are the primary Phase 1 evidence and are revisited independently by the TFT v2 attention pattern in Phase 2. The headline-bias experiment (Section 4.2.2) adds a separate finding: title-only and title-plus-body FinBERT disagree on 41.6% of articles, with titles leaning more bullish than the bodies they head, which motivates extracting features from full article bodies in Phase 2.

VAR exploration (abandoned). An exploratory VAR was fitted but abandoned: more than half of the hours carry no contemporaneous news, so the sentiment series is dominated by zeros and the impulse responses were not significant. The failure is itself diagnostic; it identifies the data as sparse and event-driven, the property that motivates the deep-learning model of Phase 2.

Limitations that drive Phase 2. Five limitations of the Phase 1 pipeline each map to a Phase 2 change (Table 4.5).

Each Phase 2 step responds to a concrete limitation exposed by the previous one, so the methodological progression is traceable rather than asserted.

Table 4.5: Phase 1 limitations and the Phase 2 responses.

Phase 1 limitation	Phase 2 response
FinBERT reduces each article to one sentiment axis (no magnitude, event type, certainty, horizon, entities)	Structured LLM extraction with a richer schema (Section 4.3.2)
The regex <code>body_valid</code> filter is fast but not semantically discriminative (false positives and negatives)	LLM-judged <code>usable</code> flag (Section 4.3.3)
The sentiment signal is sparse (~50% of hours have no news), defeating models that need joint observation	A TFT built for sparse, event-driven inputs (Section 4.3.5)
The corpus is pre-war only (ends February 2026)	Corpus extended through May 2026 (Section 4.3.1)
The OLS has no macro controls	DXY and VIX added as exogenous covariates (Section 4.3.5)

4.3 Phase 2: refined pipeline and deep-learning modeling

Phase 2 addresses the Phase 1 limitations (Section 4.2.4) through a coordinated redesign: FinBERT is replaced by structured LLM extraction, DXY and VIX enter as macro covariates, the regex filter is replaced by an LLM-judged `usable` flag, a Temporal Fusion Transformer becomes the primary model, and the corpus is extended through May 2026. The phase unfolds chronologically: an initial LLM schema (v1) supports the first TFT training (TFT v1); an inter-model calibration then exposes a weakness in the composite sentiment score, motivating a revised schema (v2) with three orthogonal economic channels and the `usable` flag; the full extraction runs on the expanded corpus and supports the final TFT v2. The subsections follow this order: data setup (Section 4.3.1), LLM extraction (Section 4.3.2), filter migration (Section 4.3.3), first calibration (Section 4.3.4), TFT v1 (Section 4.3.5), the schema response (Section 4.3.6), TFT v2 (Section 4.3.7), and a Phase 1 vs Phase 2 comparison (Section 4.3.8).

4.3.1 Phase 2 data and feature setup

The Phase 2 corpus extends Phase 1 in two directions: the GDELT scrape runs through May 2026 to cover the post-war period, and five new queries target geopolitical and demand-side themes (Section 3.1). The full corpus is **22,795 article records** (January 2024 to May 2026); **19,619 have substantive body content** (non-null, not a scraping error) and go to LLM extraction, while 3,176 returned null bodies (paywall, fetch failure, JavaScript rendering).

Macro covariates. DXY (DX-Y.NYB) and VIX ($\sqrt{\text{VIX}}$) hourly series were added as exogenous controls (Section 3.6.2). The rationale is empirical: the Phase 1 OLS explained under 0.3% of volume variance (Section 4.2.3), so macro controls let the model absorb broad market-driven variation and isolate a cleaner residual for news. The pipeline was also migrated from CSV intermediates to the SQLite database (`wti_thesis.db`), making the alignment a deterministic join.

Aligned dataset. Re-running the temporal alignment (Section 3.2) over the expanded corpus yields **22,795 article-aligned rows** (15,290 contemporaneous, 7,505 forward-assigned) on the 11,232-hour market grid spanning 13 May 2024 to 13 May 2026. Table 4.6 contrasts the two phases.

Table 4.6: Phase 1 versus Phase 2 dataset.

Stage	Phase 1	Phase 2
GDELT queries	8	13
Article window	March 2024 – February 2026	January 2024 – May 2026
Raw articles (GDELT scrape)	51,948	76,345 (cumulative; Phase 2 re-scrape added 24,397)
After deduplication + English	16,326	22,795
Articles with substantive bodies	7,756 (<code>body_valid=1</code>)	19,619 (non-null, non-error body)
Title-only fallback	5,934	0 (no title-only fallback in Phase 2)
Modeling-ready corpus	13,690 (Phase 1 OLS/VAR)	10,514 (<code>usable_strict</code>) / 11,675 (<code>usable</code>)
Market hours window	March 2024 – February 2026	May 2024 – May 2026
Hourly market observations	11,219	11,232
Macro covariates	none	DXY, VIX
Sentiment representation	FinBERT 3-class probabilities	LLM continuous sentiment + 3 channels + categorical fields

The Phase 2 dataset is the substrate for all subsequent subsections of this chapter.

4.3.2 LLM feature extraction

Replacing FinBERT with LLM extraction is the central Phase 2 change: it recovers the dimensions FinBERT’s single sentiment axis discards (magnitude, event type, certainty, time horizon, entities, and the economic channels). The methodology is specified in Sections 3.3.2 through 3.3.5; this subsection reports its Phase 2 deployment and output.

Extraction setup. Each of the 19,619 articles with a substantive body is scored by Claude Haiku 4.5 through the tool-use API, which forces the `extract_article_features`

schema (Section 3.3.3) and is parsed deterministically from the tool call; the title and body (truncated to 1,500 characters) form the user message, and the full prompt and tool schema are in Appendix A. The schema short-circuits unusable articles to only the `usable` flag, cutting output tokens for the roughly 40% unusable share, and all 19,619 articles returned a valid response with zero unrecoverable errors. The deployed schema is v2 (Section 3.3.4): the composite `sentiment_score` plus three orthogonal channels (`supply_impact`, `demand_impact`, `risk_premium`), `magnitude`, `certainty`, `event_type` (an array of one to three categories), `time_horizon`, `entities`, and the required `usable` flag. The batch filtered on non-null, non-error bodies rather than the Phase 1 regex, so the LLM’s own `usable` flag becomes the downstream filter (compared against the regex in Section 4.3.3).

Extraction outcome. The LLM flagged **11,675 articles as usable (59.5%)** and 7,944 as unusable (40.5%, predominantly paywalls, Cloudflare blocks, and keyword-matched off-topic content), consistent with the pre-batch estimate and the calibration sample (Section 4.3.4). Across the 11,675 usable articles the four continuous scores are given in Table 4.7.

Table 4.7: Continuous feature means over the 11,675 usable articles.

Feature	Mean	Std
<code>sentiment_score</code>	+0.025	0.501
<code>supply_impact</code>	−0.071	0.463
<code>demand_impact</code>	−0.075	0.340
<code>risk_premium</code>	+0.156	0.426

The near-zero `sentiment_score` mean indicates a roughly bearish/bullish-balanced corpus; the negative `supply_impact` and `demand_impact` means reflect a tilt toward supply-tightening and demand-weakening news; and the positive `risk_premium` mean (the largest in magnitude) reflects the elevated-geopolitics regime of the coverage period. A notable 9.9% of usable articles carry all three channels at exactly zero, articles the LLM judged usable but materially neutral; Section 4.3.4 revisits this, and Section 4.3.3 uses it to define the stricter `usable_strict` filter.

Categorical and entity distributions. First-rank event types are dominated by `geopolitical` (37.6%) and `supply` (28.6%), then `macro` (14.5%) and `demand` (12.7%). The most frequent entities are Iran (3,197), the United States (3,092), Russia, China, OPEC+, Trump, Israel, India, the Strait of Hormuz (832, comparable to OPEC’s count), and OPEC, all intuitive for the coverage period and evidence that the entity normalization correctly isolates chokepoints and actors rather than collapsing them. These distributions are the input substrate for the TFT models that follow.

4.3.3 LLM usable flag and filter comparison

The Phase 1 regex `body_valid` filter (Section 3.4.1) and the Phase 2 LLM `usable` flag (Section 3.4.2) are two approaches to the same problem: deciding which scraped articles carry substantive content. Both labels coexist for the 19,619 processed articles, so they can be compared directly.

Result. Table 4.8 is the 2×2 contingency table.

Table 4.8: Regex versus LLM filter agreement (19,619 articles).

	LLM usable=0	LLM usable=1	Total
regex <code>body_valid</code> =0	4,578	1,491	6,069
regex <code>body_valid</code> =1	3,366	10,184	13,550
Total	7,944	11,675	19,619

The regex accepts 69.1% of articles and the LLM 59.5%; they agree on 75.2% (14,762 of 19,619), a **Cohen’s κ of 0.47** (chance-corrected, “moderate” on the Landis-Koch convention). The two off-diagonal cells characterise the disagreement: the regex rejects but the LLM accepts **1,491** articles (regex false negatives, short but substantive content below the 400-character threshold), while the regex accepts but the LLM rejects **3,366** (regex false positives, long keyword-matched content not about WTI). The regex is permissive on long non-substantive content (it cannot judge topic semantically) and restrictive on short substantive content (its length cutoff is blunt); the LLM, with semantic access to the body, behaves the opposite way on both axes.

Manual audit. Sampling both off-diagonal cells and hand-labelling confirms the LLM’s judgment in the large majority of cases. The regex’s false positives are dominated by keyword collisions (palm oil, canola, a semiconductor firm with an energy-sounding name, generic equity wraps); its false negatives are short but on-topic briefs (for example a 320-character note on OPEC+ holding its production plan). The audit also surfaced a residual LLM error mode: a few articles the LLM marks `usable` are not genuinely WTI-relevant (the clearest being an article on Canada rolling back EV tariffs), so the LLM filter is more accurate than the regex on both of the regex’s failure modes but is not itself a ground-truth oracle, a limitation revisited in Chapter 5.

The `usable_strict` variant. The residual LLM false positives share a diagnostic pattern: the article is marked `usable` yet all three channel scores are zero, meaning the LLM’s own downstream reasoning judged no material market impact. Approximately **1,161** articles show this (the 9.9% noted in Section 4.3.2). We therefore define a stricter filter, `usable_strict`=1, requiring at least one non-zero channel; it is the canonical filter for TFT v2 training and the reported ablation (Appendix C), whereas `usable`=1 is used only for this filter comparison. It reduces training noise without imposing external judgment.

Filter choice and cross-phase note. The two filters are not nested: the LLM accepts 1,491 articles the regex rejects and rejects 3,366 it accepts, so its acceptance set overlaps but is not a subset of the regex’s. The LLM filter is the canonical Phase 2 downstream filter. This migration (plus the loss of Phase 1’s title-only fallback) means the Phase 1 modeling set (13,690) and the Phase 2 set (11,675 usable / 10,514 strict) are not directly comparable, a point developed in Section 4.3.8.

4.3.4 Inter-model calibration of LLM features

The LLM-extracted features are the substrate of all Phase 2 modeling, so they are validated before use. Following Section 3.7, we run an **inter-model calibration** [Gilardi et al., 2023, Zheng et al., 2023]: the same 30 stratified articles are scored by Claude Haiku 4.5 and by a GPT-family reference model with the identical prompt and schema, treating cross-family disagreement as a signal of genuine ambiguity or systematic bias. This subsection reports the first calibration, on the initial Schema v1, and the diagnostic it prompted.

Result on Schema v1. Schema v1 carried a single composite `sentiment_score` as its directional feature. The two models agreed on the `usable` flag for 26 of 30 articles (87%), but agreement on the composite sentiment was poor: a Pearson correlation of only **0.39**, with **opposite-signed** scores on **4 of 13 usable articles (31%)**. Sign disagreement is the most consequential kind, since it means the two models read the same article as pushing WTI price in opposite directions.

Diagnostic (the Phase 2 pivot). The sign disagreements were not random: they concentrated on high-magnitude geopolitical events, exactly the articles that move the market. The root cause is that the composite `sentiment_score` conflates two separable judgments, the **valence of the event** (is this good or bad news?) and its **directional price impact** (does it push crude up or down?). For most articles the two align, but for a supply-threatening escalation they diverge: the event is negative in valence yet bullish for price, so a valence-weighting model scores it negative and a price-weighting model scores it positive. A single number cannot encode both, so the two models resolve the ambiguity differently precisely where it matters most. A 0.39 correlation on the primary feature is structurally, not incidentally, inadequate.

Response. The fix is to decompose the judgment rather than tweak the prompt: Section 4.3.6 introduces the three orthogonal economic channels (`supply_impact`, `demand_impact`, `risk_premium`) and the re-calibration that validates them. First, though, TFT v1 was trained on the v1 schema; its results and limitations follow.

4.3.5 Temporal Fusion Transformer (TFT) v1

TFT v1 is the project’s first deep-learning model and the first to use the richer LLM feature set. It was trained on the Phase 1 corpus (Schema v1 features plus the DXY and VIX macro covariates). The Temporal Fusion Transformer [Lim et al., 2021] is an attention-based multi-horizon forecaster whose Variable Selection Network (VSN) and interpretable attention make it well suited to this analysis.

Training setup. The architecture (Section 3.6) uses a 48-hour encoder, a one-hour horizon, `hidden_size=32`, `attention_head_size=4`, `dropout=0.1`, and a quantile loss ($\sim 113,000$ parameters). On 10,797 hourly rows with an 80/20 temporal split, early stopping triggered at epoch 21 with a best validation loss of **0.204**. Inputs are the Schema v1 LLM features (`sentiment_score`, `magnitude`, integer-encoded `event_type/price_direction/time_horizon`, `certainty`, `n_articles`), the market context (`log_volume`, `price_range`, `log_return`, `amihud`, `dxy`, `vix`), and calendar covariates.

Feature importance. The VSN weight is dominated by one feature (Table 4.9).

Table 4.9: TFT v1 feature importance.

Feature	Importance	Group
<code>sentiment_score</code>	0.53	News
<code>log_volume</code>	0.095	Market
<code>dxy</code>	0.075	Market
<code>log_return</code>	0.055	Market
<code>event_type_num</code>	0.030	News
<code>magnitude</code>	0.025	News
<code>vix</code>	0.005	Market

Three points stand out. First, **`sentiment_score` dominates (0.53, over five times the next feature)**, a strong validation of the LLM approach: the continuous LLM sentiment carries far more signal than any market variable, and far more than the FinBERT label could. Second, **`VIX` is negligible (0.005)** while `DXY` ranks third, so for WTI the dollar index already absorbs most of the macro-risk signal. Third, the categorical features (`event_type`, `price_direction`, `time_horizon`) were integer-encoded, imposing a meaningless numeric order that likely understates their informativeness, a limitation fixed in TFT v2.

Attention and RQ1. The attention peaks at **lag $-4h$** (with a $-2h$ to $-5h$ cluster), independently corroborating the Phase 1 lag OLS peak at $+6h$ (Section 4.2.3): two very different methods place the dominant news-impact horizon several hours after publication. A secondary cluster at **$-27h$ to $-28h$** is a “daily memory” effect (the same hour on the

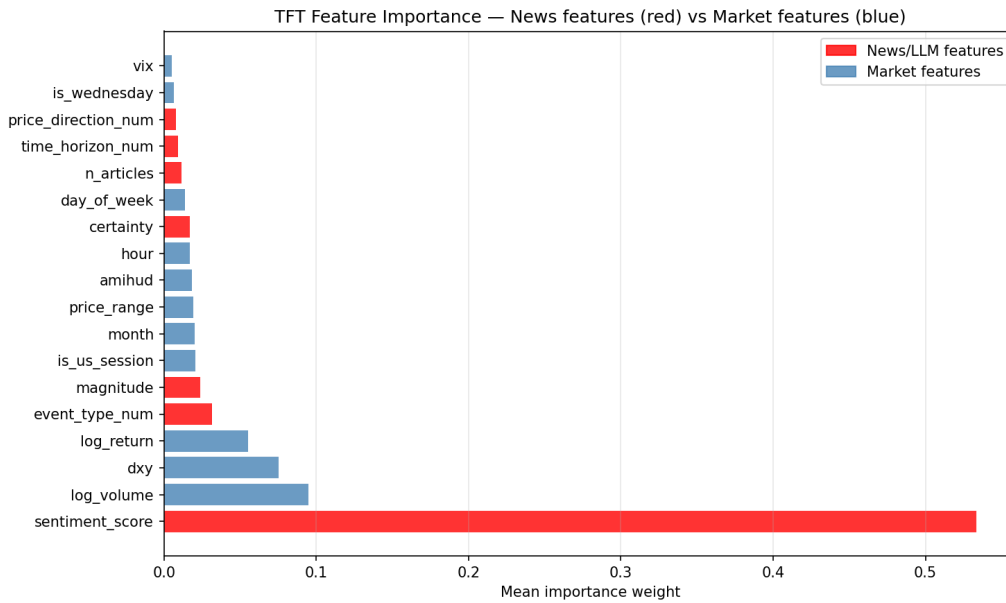


Figure 4.2: TFT v1 feature importance, news features (red) versus market features (blue). `sentiment_score` dominates at 0.53, more than five times the next feature; the market-context features form a second tier and the rest carry minor weight.

previous trading day), a structure invisible to the 12h-max lag OLS.

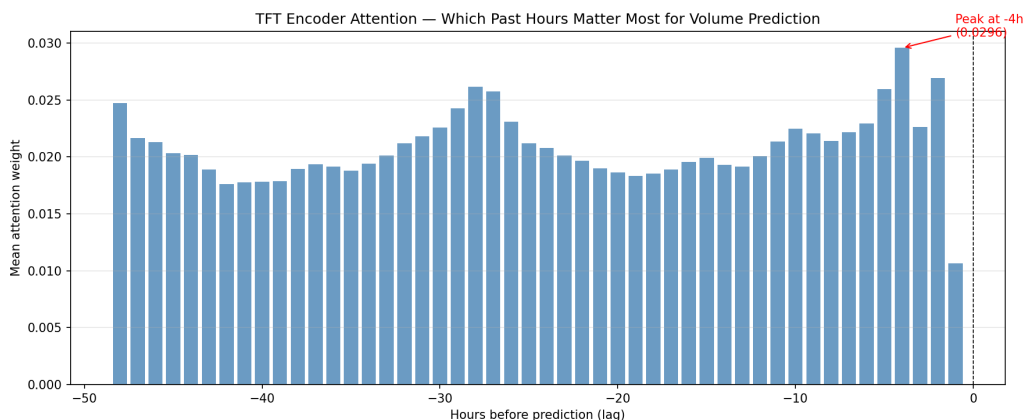


Figure 4.3: TFT v1 encoder attention by lag. Mean attention over the 48-hour encoder window, peaking at -4 h. Two clusters are visible: a short-term one between -2 h and -5 h, and a secondary one around -27 h to -28 h.

Directional asymmetry (RQ2). Splitting validation samples by sentiment direction gives the mean predicted `log_volume` in Table 4.10.

Bearish predicted volume exceeds bullish, consistent in direction with Section 4.2.3, but a two-sample t-test on the difference is not significant ($p = 0.56$), underpowered by the few hundred hours per category. The lag OLS remains the canonical RQ2 evidence; the TFT corroborates only the direction. Neutral hours carry the highest predicted volume of the three, a speculative “uncertainty drives trading” aside for the discussion.

Limitations of TFT v1 that drive TFT v2. TFT v1 validates the approach (LLM

Table 4.10: TFT v1 mean predicted `log_volume` by sentiment direction.

Sentiment direction	Mean predicted <code>log_volume</code>
Bearish	8.775
Bullish	8.700
Neutral	8.875

sentiment dominates, attention corroborates the lag OLS, and the asymmetry is reproduced in direction), but five limitations each motivate a Phase 2 change:

1. **Underpowered asymmetry test** ($p = 0.56$), motivating a larger corpus.
2. **Integer-encoded categoricals** impose a meaningless numeric order, motivating proper categorical encoding.
3. **No entity awareness** (entities extracted but unused), motivating 71 entity flags.
4. **Pre-war training data only**, motivating a corpus extended through May 2026.
5. **The dominant feature is the unreliable composite sentiment:** the 0.39-correlation feature from Section 4.3.4 carries 53% of the model’s weight, motivating the channel decomposition.

The fifth is the pivot: the model leans hardest on exactly the feature the calibration flagged as least trustworthy, so v2 replaces it with the orthogonal channels (Section 4.3.6) and retrain (Section 4.3.7).

4.3.6 Channel decomposition response

The calibration (Section 4.3.4) showed the composite `sentiment_score` was unreliable ($r = 0.39$, 31% sign disagreement) precisely on the high-magnitude geopolitical events that matter most, yet TFT v1 leaned on it for 53% of its weight. This subsection documents the schema revision that fixes this and the re-calibration that validates it.

Design principle. An oil-market article carries up to three economically distinct judgments: its implication for **supply**, for **demand**, and for **geopolitical or operational risk**. This decomposition mirrors the standard structural separation of oil-price drivers into supply, aggregate-demand, and precautionary (risk) components [Kilian, 2009]. A single composite forces the three into one number; when they diverge (a supply shock is bearish for the event but bullish for price; a de-escalation is positive in valence but bearish-supply via sanctions relief) the composite breaks down. Schema v2 therefore decomposes the judgment so each score captures one factual claim.

Schema v2 changes. The core addition is the three orthogonal channels, each on $[-1, +1]$: `supply_impact` (positive = more oil to market), `demand_impact` (positive =

stronger consumption), and `risk_premium` (positive = elevated risk). Schema v2 also promotes `usable` to a required field, makes `event_type` a salience-ordered array, drops the redundant `price_direction`, and retains the composite `sentiment_score`. The full field-by-field v1-vs-v2 comparison is in Appendix B.

Re-calibration. Re-scoring the same 30-article sample with Schema v2 improves every metric (Table 4.11).

Table 4.11: Inter-model re-calibration metrics, Schema v1 versus v2.

Metric	v1 schema	v2 schema
<code>usable</code> agreement	26/30 (87%)	27/30 (90%)
<code>sentiment_score</code> correlation	0.39	0.88
<code>sentiment_score</code> sign disagreements	4/13 usable (31%)	1/14 usable (7%)
<code>supply_impact</code> correlation	—	0.94
<code>supply_impact</code> MAD	—	0.11
<code>supply_impact</code> sign disagreements	—	0
<code>demand_impact</code> correlation	—	0.96
<code>demand_impact</code> MAD	—	0.05
<code>demand_impact</code> sign disagreements	—	0
<code>risk_premium</code> correlation	—	0.82
<code>risk_premium</code> MAD	—	0.13
<code>risk_premium</code> sign disagreements	—	1

The headline results: `sentiment_score` correlation jumps **0.39 to 0.88** and its sign disagreements drop **31% to 7%**, and each new channel correlates at 0.82 or above (`supply_impact` 0.94, `demand_impact` 0.96, `risk_premium` 0.82) with near-zero sign disagreements. Strikingly, the composite improved even though it was not modified: asking the model to decompose the article first, and only then produce a composite, disciplines every output it touches. The channels are genuinely orthogonal (within-model pairwise $|r| < 0.5$), so they capture distinct economic dimensions rather than one relabelled judgment.

Residual weakness. On 3 of the 14 usable calibration articles, Haiku zeroed all three channels where GPT assigned small non-zero values, on articles of modest relevance; the corresponding `sentiment_score` and `magnitude` were also subdued, so these do not dominate downstream, but this conservatism is the same pattern that defines the `usable_strict` filter (Section 4.3.3). With the channels validated, the full Schema v2 batch (19,619 articles) produces the feature set for TFT v2 (Section 4.3.7).

4.3.7 Temporal Fusion Transformer v2

TFT v1 validated the deep-learning approach; TFT v2 integrates the Phase 2 investments (channel decomposition Section 4.3.6, LLM filter Section 4.3.3, expanded corpus, entity

normalization) and is the model against which the research questions are evaluated. A controlled ablation guided the design (Appendix C); the reported configuration is the one selected on validation performance.

4.3.7.1 Success criteria

Three criteria reflect Phase 2’s complementary role: the Phase 1 OLS provides the primary evidence for RQ1 (+6h peak) and RQ2 (bearish > bullish), and TFT v2 is expected to corroborate this through an independent method while showing the channels and entities as visible drivers.

1. **Channels and entities are visible drivers.** The VSN importance ranking should place channel features and entity flags in the top ranks, with a non-degenerate, interpretable attention pattern.
2. **Prediction substantially beats persistence** in the 1-to-12h window, confirming the Phase 2 features carry predictive signal.
3. **The multi-horizon error curve is consistent with Phase 1’s lag structure**, with the largest persistence reduction in the +6h range identified by the lag OLS.

The evaluation is in Section 4.3.7.7.

4.3.7.2 Model design

TFT v2 uses a hidden size of 32, four attention heads, dropout 0.15, and a hidden-continuous size of 16 (298,329 parameters), trained on the `usable_strict=1` filter (Section 4.3.3, which drops ~1,161 channel-neutral articles labelled as `usable`). The input features are:

- **Channels (Phase 2 contribution):** `supply_impact`, `demand_impact`, `risk_premium`, each on $[-1, +1]$.
- **Composite news:** `sentiment_score`, `magnitude`, `certainty`, `n_articles`.
- **Categoricals:** `event_type_primary` (8 categories including the synthetic `no_news`) and `time_horizon` (4), as proper embeddings (dimension 5 and 3), taking the dominant article’s value (highest `magnitude`, tie-broken on `article_id`).
- **Entity flags (Phase 2 contribution):** 71 binary multi-hot columns (Section 4.3.2).
- **Market context:** `log_volume`, `price_range`, `log_return`, `amihud`, `dxy`, `vix`.
- **Calendar:** `hour`, `day_of_week`, `month`, `is_us_session`, `is_wednesday`.

Continuous features are hour-averaged, entity flags use the maximum, and categoricals take the highest-magnitude article; news-free hours get the `no_news` category and zeros.

A handful of boundary nulls (DXY/VIX holiday gaps, first-hour `log_return/amihud`) are forward-filled or set to zero, affecting under 0.1% of samples.

The model predicts three targets jointly (`log_volume`, `amihud`, `price_range`) at four horizons (**1, 3, 6, 12h**); the +6h horizon matches the lag OLS peak (Section 4.2.3). The temporal split is **60/20/20** on the 11,232 hourly observations, with 48h buffers between partitions: train 2024-05-13 to 2025-06-11 (6,728 samples), validation 2025-06-13 to 2025-12-27 (2,216), test 2025-12-29 to 2026-05-13 (2,159). The war onset (28 February 2026) falls inside the test window (994 pre-war, 1,165 war hours), so the test measures generalization from pre-war training to the unseen war regime; metrics are reported on the full test set, the pre-war slice, and the war slice throughout.

Training uses the Adam optimizer (learning rate 1e-3, halved after three epochs without validation improvement). The objective is a quantile loss applied independently to each of the three targets and summed, so the model learns to predict a spread of quantiles (a prediction interval) per target rather than a single point estimate. Each target is normalized separately, with the normalization statistics pooled within the asset group (a single group here, WTI), so the three targets are placed on comparable scales before their losses are combined. Gradients are clipped at 0.1, training stops early once validation loss stops improving (patience 10, minimum delta 1e-4), and the run is seeded (42) with deterministic algorithms on a Colab T4 GPU. The best validation loss of **0.427** was reached at epoch 10.

4.3.7.3 Predictive performance

We evaluate TFT v2 on the held-out test set for all three targets at the four horizons, against a persistence baseline (the current value carried forward, a standard financial-forecasting reference), reporting the median quantile (q50) on four slices: validation, full test, the pre-war test portion (994 hours before the 28 February 2026 onset), and the war portion (1,165 hours from onset to the end of the data).

Table 4.12: `log_volume`: TFT v2 versus persistence MAE.

Horizon	Persistence MAE	TFT v2 MAE	Reduction	Pre-war MAE	War MAE
1h	1.076	0.585	46%	0.536	0.628
3h	1.452	0.577	60%	0.537	0.611
6h	1.820	0.602	67%	0.551	0.646
12h	2.174	0.631	71%	0.568	0.685

The error reduction over persistence grows monotonically with horizon, from 46% at 1h to a peak of 71% at 12h, as persistence weakens at longer horizons (volume autocorrelation decays); the +6h/+12h peak matches the lag OLS range of Section 4.2.3. Error is 17

to 21% higher on the war slice than pre-war at every horizon, the expected cost of extrapolating from pre-war-only training, but modest enough that the model retains substantial predictive value on the unseen regime.

Table 4.13: `amihud`: TFT v2 versus persistence MAE.

Horizon	Persistence MAE	TFT v2 MAE	Reduction	Pre-war MAE	War MAE
1h	0.0004	0.00023	43%	0.00010	0.00035
3h	0.0004	0.00023	43%	0.00010	0.00034
6h	0.0004	0.00022	45%	0.00009	0.00034
12h	0.0004	0.00022	45%	0.00009	0.00033

A steady 43 to 45% reduction across horizons, with little horizon variation (partly an artifact of `amihud`’s compressed $1e-4$ scale). War-slice persistence MAE is three to four times the pre-war value, reflecting illiquidity spikes in the war regime, yet the model’s relative reduction over persistence holds on both slices.

Table 4.14: `price_range`: TFT v2 versus persistence MAE.

Horizon	Persistence MAE	TFT v2 MAE	Reduction	Pre-war MAE	War MAE
1h	0.495	0.718	−45%	0.180	1.200
3h	0.578	0.719	−24%	0.180	1.200
6h	0.630	0.721	−14%	0.180	1.198
12h	0.701	0.724	−3%	0.180	1.201

Here the model underperforms persistence on the full test set (−3 to −45%), a regime-specific failure: on the pre-war slice it matches persistence (MAE approximately 0.180), but on the war slice it degrades to approximately 1.200, worse than war-slice persistence (0.764 to 1.059). Trained only on the moderate-volatility pre-war regime, the model reverts toward the historical mean when forced to predict the unseen high-volatility war regime, whereas persistence at least tracks the elevated current state. This is a clean regime-extrapolation failure, and `price_range`, a direct intraday-volatility measure, is far more regime-sensitive than the aggregate targets; we return to the implications in Chapter 5.

Across targets. The model establishes substantial gains on `log_volume` (46 to 71%) and `amihud` (43 to 45%) but fails on `price_range` in the war regime, consistent with the train/test regime mismatch: the aggregate liquidity measures transfer across regimes, whereas `price_range` does not. The remainder of Section 4.3.7 therefore focuses on `log_volume`, the target most relevant to RQ1 and RQ2 and the one on which the model is strongest; the `price_range` failure mode recurs in Section 4.3.7.5 and Chapter 5.

4.3.7.4 Feature contributions

The Variable Selection Network (VSN) assigns each input feature a learned weight reflecting how much the model relies on it; aggregated over validation samples, these weights rank the features. Table 4.15 reports the top ten for `log_volume` prediction.

Table 4.15: TFT v2 top ten features by VSN importance (`log_volume`).

Rank	Feature	Importance	Type
1	<code>vix</code>	0.188	macro covariate
2	<code>supply_impact</code>	0.121	channel (news)
3	<code>ent_oman</code>	0.113	entity flag (news)
4	<code>demand_impact</code>	0.055	channel (news)
5	<code>is_wednesday</code>	0.022	calendar
6	<code>ent_japan</code>	0.017	entity flag (news)
7	<code>ent_eu</code>	0.016	entity flag (news)
8	<code>ent_iran</code>	0.015	entity flag (news)
9	<code>ent_china</code>	0.014	entity flag (news)
10	<code>ent_algeria</code>	0.014	entity flag (news)

Three patterns stand out. First, VIX ranks first at 18.8%, consistent with its role as a market-wide volatility proxy: `log_volume` tracks broad equity-market activity through stress periods, and risk expectations modulate the volume response to oil news. Second, the channel decomposition places two features in the top five, `supply_impact` (12.1%) and `demand_impact` (5.5%), together 17.6% of importance and comparable to VIX alone; the composite `sentiment_score` does not appear in the top ten, indicating that once the model has the decomposed channels the aggregate sentiment adds little marginal information. Third, six of the top ten are entity flags, validating the entity normalization of Section 4.3.2, and they are economically interpretable: `ent_oman` (a producer adjacent to the Strait of Hormuz), `ent_iran` (the actor at the center of the conflict), `ent_japan`/`ent_eu`/`ent_china` (major importers), and `ent_algeria` (a mid-size OPEC+ producer).

The two feature-engineering choices behind this result, proper categorical encoding and the entity flags, were validated in a controlled three-variant ablation whose key finding is that the channels’ predictive role emerges only under proper categorical encoding (where `demand_impact` becomes the top feature), not under the integer encoding of the simplest variant, in which the composite sentiment score dominates instead. The full three-variant comparison is deferred to Appendix C to keep the main text focused on the canonical model.

The VSN reports a feature’s weight, not the direction or magnitude of its effect, so this interpretation is qualitative: VIX informs volatility expectations, the channels capture fundamental-driven supply and demand news, and the entity flags carry source-specific

risk signals. Sections 4.3.7.5 and 4.3.7.6 examine how the model turns these features into predictions, through its attention pattern and per-horizon error structure.

Stability of the ranking. The specific ordering in Table 4.15 is not fully robust to retraining. An independent run, retrained after adding two candidate covariates and a minor data-cleaning fix, reshuffled the ranking substantially: `risk_premium` and `magnitude` rose to the top two positions while `supply_impact` and `ent_oman` fell out of it. What is stable across both runs is the character of the leading features rather than their exact order: risk and volatility proxies (VIX, `risk_premium`) and news-salience measures dominate in both, while the composite `sentiment_score` and the article count stay near the bottom. The importance results should therefore be read at the level of feature type, risk and salience over price direction (Section 4.3.7.6 and Chapter 5), rather than as a precise per-feature ranking.

4.3.7.5 Lag structure analysis: evidence for RQ1

Research Question 1 asks at what temporal lag news events most strongly affect liquidity. Phase 1’s lag OLS (Section 4.2.3) placed the bearish-news peak on `log_volume` at +6h, with a secondary trace at +12h. We test TFT v2 against this through two complementary diagnostics: the per-horizon error curve and the attention pattern.

Per-horizon error curve. If news impact concentrates at a specific horizon, the multi-horizon model should be most accurate there after normalizing for baseline difficulty, which the MAE-reduction-over-persistence of the `log_volume` curve (Table 4.12) does. That curve grows monotonically from 46% at +1h to 71% at +12h, placing the strongest persistence-relative improvement in the +6h to +12h window, the same range lag OLS identified (the exact peak differs slightly: +12h for TFT v2, +6h for lag OLS). The two methods answer structurally different questions, lag OLS isolates the direct effect of a single sentiment value at lag $-k$, whereas the TFT predicts forward from the full 48-hour window and all features jointly, so their agreement on this window is a substantive corroboration through an independent methodology rather than a restatement of one finding.

Attention pattern. The encoder attention distribution, aggregated over validation samples, gives a complementary view of the lag structure.

Figure 4.4 shows mean encoder attention across the 48-hour window. Overall attention peaks at lag -1 h (weight 0.0315) and falls off smoothly through -2 h to -5 h (0.0308, 0.0304, 0.0301, 0.0300), a graded recency effect with no boundary concentration. Disaggregated by sentiment direction the pattern diverges: bearish-sentiment hours peak at -6 h (0.0321) while bullish-sentiment hours peak at -1 h (0.0347). The bearish peak at -6 h aligns exactly with the +6h lag OLS peak, indicating the model has learned that bearish news

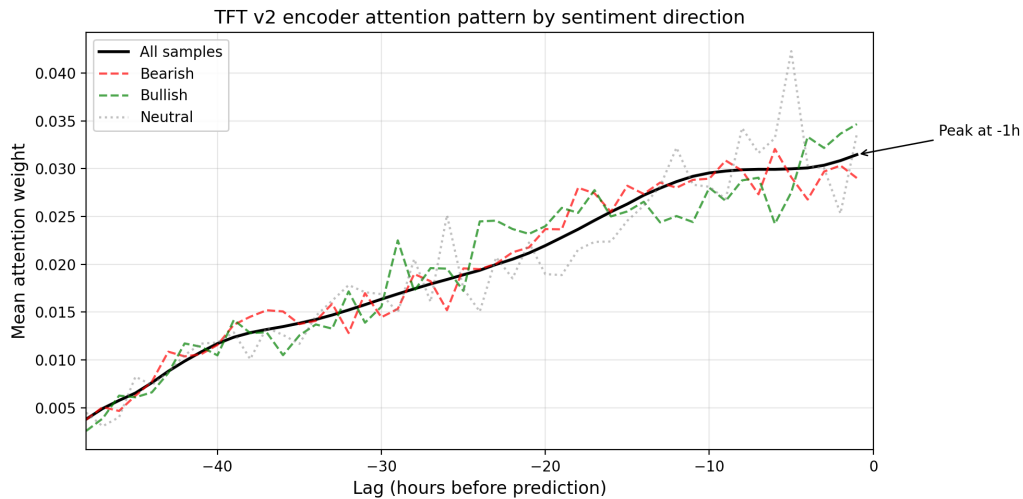


Figure 4.4: TFT v2 mean encoder attention across the 48-hour window, disaggregated by sentiment direction. Overall attention peaks at $-1h$; bearish-sentiment hours peak at $-6h$ while bullish-sentiment hours peak at $-1h$.

requires a longer integration window than bullish news, and independently confirming the same structure Phase 1 found in the average bearish impact.

Synthesis on RQ1. Both diagnostics corroborate the lag OLS finding through methodologically distinct views: the per-horizon error curve peaks in the $+6h$ to $+12h$ range, and the bearish-sentiment attention peaks precisely at $-6h$. Together they give the $+6h$ lag substantive empirical support from two independent analyses.

4.3.7.6 Directional asymmetry analysis: evidence for RQ2

Research Question 2 asks whether bearish news produces a different liquidity response than comparable bullish news. Phase 1’s lag OLS (Section 4.2.3) found a robust bearish $>$ bullish asymmetry in $\log_volume$, most pronounced at $+6h$, and remains the primary answer to RQ2; here we test whether TFT v2’s predictions provide complementary evidence through an independent method.

Test design. For each horizon and slice we split samples by the sentiment score at the prediction hour, bearish (below -0.1), bullish (above $+0.1$), and neutral (within plus or minus 0.1), and test whether the mean predicted $\log_volume$ differs between the bearish and bullish subsets with Welch’s t-test (unequal variance). Neutral hours are excluded: they capture low-news activity, dominate the sample counts, and dilute the contrast the test is designed to detect. This yields sixteen tests (four horizons times four slices: val, test_full, test_prewar, test_war), reported in Table 4.16.

Results. None of the sixteen tests reaches significance at $p < 0.05$. The closest is $+12h$ on test_war (bearish 8.169 vs bullish 8.540, difference -0.371 , $p = 0.053$); two others approach without crossing, $+3h$ test_prewar (bearish $>$ bullish, $+0.341$, $p = 0.058$) and

Table 4.16: Directional asymmetry tests: mean predicted `log_volume`, bearish versus bullish.

Horizon	Slice	Bear. n	Bull. n	Bear. mean	Bull. mean	Diff.	p	Sig.
1h	val	320	271	8.345	8.281	+0.064	0.578	No
1h	test_full	384	826	8.928	8.857	+0.071	0.316	No
1h	test_prewar	180	340	8.853	8.783	+0.070	0.543	No
1h	test_war	204	486	9.001	8.901	+0.100	0.253	No
3h	val	320	271	7.938	8.131	-0.193	0.270	No
3h	test_full	384	826	8.732	8.596	+0.136	0.247	No
3h	test_prewar	180	340	8.706	8.365	+0.341	0.058	No
3h	test_war	204	486	8.756	8.729	+0.027	0.869	No
6h	val	320	271	8.186	8.369	-0.183	0.230	No
6h	test_full	384	826	8.633	8.529	+0.104	0.400	No
6h	test_prewar	180	340	8.470	8.380	+0.089	0.636	No
6h	test_war	204	486	8.788	8.615	+0.174	0.285	No
12h	val	320	271	8.277	8.371	-0.094	0.561	No
12h	test_full	384	826	8.220	8.464	-0.244	0.070	No
12h	test_prewar	180	340	8.274	8.333	-0.059	0.761	No
12h	test_war	204	486	8.169	8.540	-0.371	0.053	No

+12h test_full (bearish < bullish, -0.244 , $p = 0.070$).

Interpretation. This test and Phase 1’s lag OLS estimate different quantities, and the distinction governs how the null should be read. The OLS compares the marginal coefficients on bearish and bullish sentiment (the sensitivity of volume to each direction holding the other fixed, Section 4.2.3). The test here compares the mean predicted volume of bearish hours against bullish hours, an unconditional difference of group means with nothing held fixed. A larger marginal sensitivity to bearish news, Phase 1’s finding, can coexist with near-equal average volume across bearish and bullish hours (for instance if bullish hours cluster in higher-baseline-volume periods, such as risk-on sessions), so the absence of a group-mean asymmetry here does not contradict Phase 1; it answers a coarser, confounded question.

The point-prediction differences are small and non-significant at every horizon, with no consistent sign across slices. This is not a power artifact: the bearish and bullish groups each hold hundreds of samples, and no test becomes significant under alternative sentiment thresholds (swept from 0.0 to 0.5) once corrected for multiple comparisons. It instead mirrors the target itself. In the actual test data, the future volume of bearish and bullish hours is statistically indistinguishable at the 1h to 6h horizons, and the only significant ground-truth difference, at +12h, runs bullish above bearish (a sign the model reproduces, predicted difference -0.244). The model is therefore tracking a group-level near-symmetry in its target period, not washing out a signal that is present. The one directional structure the TFT does carry is temporal rather than in magnitude: the attention pattern (Section 4.3.7.5) attends to $-6h$ for bearish sentiment and $-1h$ for

bullish, matching Phase 1’s +6h peak.

Synthesis on RQ2. Phase 1’s lag OLS remains the primary and appropriate evidence for RQ2, because it measures the directional sensitivity that the research question is about. TFT v2 shows no magnitude asymmetry in its point predictions, but this is expected rather than a failed replication: the group-mean comparison is a different, confounded estimand, and the model’s target is itself near-symmetric across sentiment direction over the test period. The model’s genuine complementary contribution is temporal, the attention asymmetry of Section 4.3.7.5 (bearish integrated over a longer window, $-6h$, than bullish, $-1h$), which corroborates the +6h lag structure of RQ1 without depending on the magnitude test.

4.3.7.7 Success criteria evaluation

We now evaluate TFT v2 against the three success criteria declared in Section 4.3.7.1, using the evidence presented in Sections 4.3.7.3 through 4.3.7.6.

Table 4.17: TFT v2 success-criteria evaluation.

Criterion	Status	Evidence
1. Channels and entities as visible drivers	Pass	<code>supply_impact</code> and <code>demand_impact</code> appear in the top four features (12.1% and 5.5%); six entity flags in the top ten; attention layer non-degenerate with peak at $-1h$; bearish/bullish attention divergence ($-6h$ vs $-1h$) matches Phase 1’s identified lag
2. Substantial persistence-relative improvement	Pass	<code>log_volume</code> MAE reduction of 46–71% across the 1–12h window and <code>amihud</code> 43–45% (<code>price_range</code> is the exception on the war regime, Section 4.3.7.3)
3. Multi-horizon curve consistent with Phase 1	Pass	Persistence reduction peaks at +12h (71%) with +6h close behind (67%); attention pattern independently confirms the +6h finding via the bearish-sentiment attention peak

TFT v2 passes all three criteria: the Phase 2 features (channels, entities) are economically interpretable and drive prediction, the model substantially beats persistence on the aggregate liquidity targets, and its multi-horizon error curve and attention independently corroborate Phase 1’s +6h lag structure. This does not make Phase 2 the source of evidence for the research questions. Phase 1’s lag OLS remains the primary statistical evidence for RQ1 (peak at +6h) and RQ2 (the bearish $>$ bullish sensitivity), while TFT v2 contributes complementary, methodologically distinct evidence: interpretable feature importance and a bearish-sentiment attention peak at $-6h$ that confirms the same lag structure. The relationship between the two phases is complementary, not competitive. The configuration evaluated here is designated TFT v2 for the remainder of the thesis.

4.3.8 Methodological comparison: Phase 1 versus Phase 2

The two phases approach the same research questions through structurally different methodologies: Phase 1 (Section 4.2) uses regression (contemporaneous and lag OLS) on FinBERT sentiment over a regex-filtered corpus, Phase 2 (Section 4.3) uses a deep-learning forecaster (TFT v2) on LLM-derived channel-decomposed features over an LLM-filtered corpus. Neither supersedes the other; they answer complementary questions about the same data.

4.3.8.1 What Phase 2 changed methodologically

Phase 2 changed the analytical stack at five levels (corpus, feature extraction, tone versus price impact, entity normalization, and analytical framework), each motivated by a specific Phase 1 limitation.

Corpus expansion. Phase 1’s 13,690-article regex-filtered corpus (`body_valid`) covers the pre-war period to February 2026. Phase 2’s LLM-filtered corpus (`usable_strict=1`, approximately 10,514 articles) extends through May 2026 to include the 28 February 2026 war onset and its aftermath; the `usable_strict` filter (Section 4.3.3) drops 1,161 topical-but-channel-neutral articles. The Phase 2 corpus is smaller in count but temporally more complete, covering the regime change Phase 1’s data does not.

Feature extraction. Phase 1’s FinBERT produces one three-class sentiment label per article (bearish, neutral, bullish), averaged hourly. Phase 2’s LLM extracts a structured schema per article: a composite sentiment score, three channels (`supply_impact`, `demand_impact`, `risk_premium`), magnitude, certainty, event type, time horizon, and entities. The channels let the model separate supply, demand, and risk-premium news that FinBERT collapses onto a single axis.

Tone versus price impact. The two sentiment signals do not only differ in resolution; they measure different constructs, and the difference is diagnostic. FinBERT classifies the surface financial tone of an article, whereas the Haiku `sentiment_score` is prompted for the net directional impact on the WTI price (“assess the actual content, not the headline tone”). Across the 6,300 articles scored by both methods the two agree only moderately (Pearson 0.54), and they diverge in one specific place: geopolitical news, where the mean FinBERT tone is negative (−0.116) while the mean Haiku price impact is bullish (+0.137); on demand, macro, and supply articles the two agree in sign. The mechanism is visible in the headlines they disagree on, for example “Russian Sanctions Drive Oil Prices Higher” (FinBERT −0.95, Haiku +0.85) and “Oil prices soar after Israel launches strikes on Iran capital” (FinBERT −0.96, Haiku +0.80): FinBERT anchors on the alarming vocabulary (sanctions, strikes) and labels the article negative even when the headline states that prices rose, because it has no asset grounding and no causal step from a supply threat to a

higher price. The LLM, prompted to reason about the price impact for WTI specifically, resolves the same articles as bullish. This is the concrete form of the FinBERT limitation that motivated the Phase 2 extraction, and it is confined to exactly the geopolitical news that carries the highest risk premium and the highest trading volume (Section 4.3.7.4). It also explains how a sentiment-sign asymmetry test can diverge between the two phases (Section 4.3.7.6): the high-volume geopolitical news falls in opposite sentiment buckets under the two definitions.

Entity normalization. Phase 1 uses no entity features. Phase 2 maps entity mentions to 71 canonical flags (Section 4.3.2 and Appendix D), giving each hour a binary vector of which actors (US, Iran, Saudi Arabia, OPEC+, etc.) the dominant article named, a signal no Phase 1 method had.

Analytical framework. Phase 1’s regression isolates the effect of specific variables at specific lags but produces no multi-horizon forecast. Phase 2’s TFT predicts three targets at four horizons jointly from the previous 48 hours, but does not isolate individual variable effects the way OLS does. The move is not from a weaker method to a stronger one but from one that isolates effects to one that integrates them, each complementing the other.

4.3.8.2 Why cross-phase numbers are not comparable

TFT v1 and TFT v2 loss values cannot be compared directly, because almost everything beneath them differs: the loss function (v1 `QuantileLoss` on one target; v2 `MultiLoss` on three, roughly three times the per-target scale), the validation period (v1 approximately 2,160 pre-war hours from an 80/20 split; v2 approximately 2,216 transitional hours from the 60/20/20 split), the horizons (v1 one hour; v2 four jointly), the feature count (v1 approximately 12; v2 over 90), and the upstream corpus and extraction (FinBERT on the regex corpus versus LLM channels on the `usable_strict` corpus). Comparing v1’s `val_loss` of 0.204 to v2’s 0.427 would therefore be misleading.

The fair comparison is persistence-relative reduction on a shared target and horizon (`log_volume` at 1h), where both models substantially beat persistence and neither fails. Phase 2’s contribution is not measurably better forecasts but a richer analytical vocabulary: where v1 concentrates importance on a scalar sentiment score, v2 distributes it across decomposed channels, entity flags, and macro covariates. The progression from Phase 1 to Phase 2 is real but qualitative and analytical, not a numerical win; the thesis’ primary contribution is the Phase 2 methodology, not a demonstration that it forecasts better.

4.4 Summary

Chapter 4 has reported the empirical results of both methodological phases of this thesis.

Phase 1 (Section 4.2) established the initial evidence for the research questions through regression-based analysis of a pre-war corpus of 13,690 articles processed with FinBERT sentiment features. The headline bias experiment (Section 4.2.2) demonstrated that title-only and title-plus-body FinBERT inputs produce substantially different sentiment classifications, motivating the shift toward richer feature extraction in Phase 2. The lag OLS (Section 4.2.3), including the contemporaneous $k = 0$ case, confirmed a positive news-volume association and identified the +6 hour lag as the peak of the news impact on liquidity, with a statistically significant bearish > bullish directional asymmetry. An exploratory vector autoregression (Section 4.2.4) was abandoned once the sparsity of the hourly sentiment signal proved incompatible with its assumptions, which reinforced the move to the deep-learning models of Phase 2.

Phase 2 (Section 4.3) extended the analysis through LLM-based feature extraction on an expanded corpus covering the pre-war and war regimes. The Schema v2 extraction produced channel-decomposed features (Section 4.3.2) that distinguish supply-side, demand-side, and risk-premium news impacts, alongside 71 canonical entity flags derived from a normalized entity list. The comparison between the regex-based Phase 1 filter and the LLM-based Phase 2 filter (Section 4.3.3) identified specific error modes in each and motivated the stricter `usable_strict=1` variant used for TFT v2 training. TFT v1 (Section 4.3.5) validated the deep-learning approach on a single target and horizon, achieving substantial improvements over persistence and identifying a two-cluster attention pattern (short-term at -4h, daily memory at -27/-28h). TFT v2 (Section 4.3.7) integrated the full Phase 2 methodological investments: channel decomposition, entity flags, proper categorical encoding, multi-horizon prediction, and multi-target learning. The v2 canonical configuration reduces prediction error over persistence by 46 to 71 percent on `log_volume` across horizons, with peak reduction at +12h consistent with the +6 to +12 hour range identified by Phase 1's lag analysis. The Variable Selection Network's feature importance ranking places VIX and `supply_impact` as the top two features and includes six entity flags in the top ten. The directional asymmetry test (sixteen bearish-versus-bullish comparisons of predicted volume) returned no significant group-level difference; this reflects a different, coarser estimand than Phase 1's marginal-coefficient asymmetry rather than a failed replication, and the TFT's complementary evidence for RQ2 is temporal, its attention attending to -6h for bearish sentiment against -1h for bullish.

The methodological comparison of the two phases (Section 4.3.8) argued that direct numerical comparison between v1 and v2 is not appropriate given differences in validation sets, loss functions, prediction horizons, feature counts, and news corpus. Both models substantially beat persistence in their respective settings; Phase 2's contribution is analytical richness rather than measurably better raw predictive accuracy. That section also established that FinBERT's tone-based sentiment and the LLM's price-impact sentiment

measure different constructs, agreeing overall but diverging in sign on the high-volume geopolitical news.

The success criteria established for TFT v2 in advance of training were evaluated in Section 4.3.7.7, and all three pass. The evaluation refines what the Phase 1 findings mean through a different methodological lens and exposes the `price_range` regime-extrapolation limitation that informs the discussion in Chapter 5. Chapter 5 develops the implications of these findings, together with the limitations and directions for future work. Chapter 6 concludes.

5. Discussion

This chapter interprets the empirical findings of Chapter 4, situates them against the prior work of Chapter 2, and states the limitations of the study. The discussion is organized around the two research questions (the lag structure of the news-liquidity response, RQ1, and its directional asymmetry, RQ2), followed by the methodological contributions of the two-phase design and the limitations that qualify the conclusions.

5.1 Lag structure of the news-liquidity response (RQ1)

RQ1 asks at what temporal lag news events exert their strongest effect on WTI liquidity. Two distinct analyses converge on the same answer.

- **Phase 1’s lag OLS** (Section 4.2.3) places the peak of the bearish news effect on `log_volume` at +6 hours, with a secondary trace at +12 hours.
- **Phase 2’s TFT v2** corroborates this through two independent diagnostics (Section 4.3.7.5): the per-horizon error curve improves most over persistence in the +6 to +12 hour window, and the encoder attention, disaggregated by sentiment direction, peaks at −6 hours for bearish-sentiment hours, aligning with the lag OLS peak.

The fact that a regression coefficient estimated on a single sentiment value at a fixed lag and a deep-learning forecaster integrating a 48-hour window of many features independently identify the same window (+6 to +12 hours) is the strongest evidence the thesis offers for RQ1.

The substantive reading is that news information does not reprice liquidity instantaneously but propagates over a multi-hour window. A purely contemporaneous response would place the peak at lag zero, and a market that fully and immediately absorbed news would show no predictable structure at +6 to +12 hours.

Instead, the response builds and persists across the trading session and into the next hour, consistent with gradual information diffusion and staggered trader reaction rather than a single immediate repricing. The TFT attention lands this picture: a strong recency component at −1 hour (the most recent news is always the most informative for the immediate forecast) coexists with a longer integration window for bearish news specifically.

The market appears to **take longer** to work through news that implies a *downward* price move than news that implies an *upward* one, a directional nuance in the temporal structure that Phase 1’s average-lag estimate cannot resolve but that the TFT’s direction-conditioned attention makes visible.

This lag structure is consistent with the premise established in the literature (Section 2.2) that news sentiment moves trading activity and not only prices [Tetlock, 2007]. The contribution here is one of resolution: prior work operating at **daily** frequency cannot observe an intraday +6 to +12 hour propagation window, which the **hourly** design of this thesis is built to detect. That news effects concentrate at high frequency, largely within hours of release, is itself established in the literature [Ederington and Lee, 1993]. The finding also frames the result in practical terms: an hourly-resolution monitor of WTI-relevant news carries information about liquidity that is most actionable not in the same hour but over the following half to full trading day.

5.2 Directional asymmetry of the response (RQ2)

RQ2 asks whether bearish news produces a different liquidity response than comparable bullish news. Phase 1’s lag OLS (Section 4.2.3) answers yes: the coefficient on bearish sentiment exceeds the coefficient on bullish sentiment at the dominant lags, a robust directional asymmetry that is the thesis’ primary evidence for RQ2. This is a statement about marginal sensitivity, how much an increment of bearish rather than bullish sentiment moves volume, holding the other fixed.

Phase 2’s TFT v2 does not reproduce this asymmetry in its point predictions: across the sixteen bearish-versus-bullish comparisons of predicted volume (Section 4.3.7.6), none is significant. It would be a mistake to read this as a failed replication or as an underpowered test, because the two analyses estimate different quantities.

The lag OLS compares marginal coefficients, a slope holding other variables fixed; the TFT test compares the average predicted volume of bearish against bullish hours, an unconditional difference of group means. A larger marginal sensitivity to bearish news can coexist with near-equal average volume across bearish and bullish hours, for instance when bullish hours cluster in higher-baseline-volume periods. The null is not a power artifact: the two groups each hold hundreds of samples, the result is robust to the sentiment threshold across the range from 0.0 to 0.5, and the target itself is near-symmetric, since in the actual test data the future volume of bearish and bullish hours is statistically indistinguishable at the 1 to 6 hour horizons. The model even reproduces the one genuine difference, a bullish-above-bearish gap at +12 hours, faithfully. It is tracking a near-symmetric ground truth, not washing out a signal that is present.

A second and deeper reason the two phases point differently is that they define sentiment differently (Section 4.3.8). FinBERT scores the surface financial tone of an article, whereas the Haiku `sentiment_score` is prompted for the net directional impact on the WTI price. The two agree only moderately (Pearson 0.54) and diverge in sign on exactly one kind of news: geopolitical events.

A headline such as “Russian Sanctions Drive Oil Prices Higher” is negative in tone (sanctions, conflict) but bullish in price (a supply threat lifts crude), so FinBERT labels it bearish while Haiku labels it bullish.

This geopolitical news is precisely the high-volume, high-risk-premium subset that dominates the liquidity response. Under Phase 1’s tone-based definition it falls in the bearish bucket, reinforcing the bearish over bullish asymmetry; under Phase 2’s price-based definition it falls in the bullish bucket, which is why the TFT’s price-conditioned test finds no bearish over bullish gap and even a mild reversal. The apparent tension between the phases is therefore not a contradiction but a consequence of measuring bearishness two different ways.

For the TFT it is not a matter of negative versus positive news, but of the *risk premium* the news implies.

What the TFT actually relies on clarifies this even more. Its most important features (Section 4.3.7.4) are VIX, the supply and demand channels, and a set of geopolitical entity flags dominated by actors around the Strait of Hormuz (Oman, Iran, Qatar). None of these is a bearish signal in the price sense; they are markers of risk and salience. The entities the model weights most are neutral-to-bullish in price impact but flag hours of elevated volume and risk premium regardless of direction. The asymmetry the model has learned is therefore not bearish over bullish but risk-and-salience over baseline: WTI liquidity responds to how much a piece of news raises uncertainty about supply, not to whether it implies a higher or lower price.

The answer to RQ2 is consequently conditional on how sentiment is defined and on which quantity is estimated. There is a robust bearish over bullish asymmetry in the marginal sensitivity of volume to negative-tone news (Phase 1), and this remains the thesis’ primary answer. It does not show as a difference in the level of predicted volume between price-bearish and price-bullish hours (Phase 2), because the volume-driving geopolitical news is price-bullish and because the model organizes its predictions around risk and salience rather than price direction. Both readings are correct within their own framing, and stating the asymmetry precisely requires naming the sentiment construct and the estimand it refers to.

5.3 Methodological contributions

Beyond its answers to the two research questions, the thesis makes four methodological contributions, each arising from a concrete problem encountered in the work rather than proposed in the abstract.

Channel decomposition as a validation substrate. The decomposition of sentiment into orthogonal supply, demand, and risk-premium channels (Section 3.3.4), grounded in the structural decomposition of oil-price shocks [Kilian, 2009], was introduced to solve a specific failure: the composite `sentiment_score` disagreed across model families (Haiku versus GPT-5.5) on high-magnitude geopolitical events (Section 4.3.4). Recalibrating on the channels raised cross-model agreement on the composite from 0.39 to 0.88, with channel correlations of 0.94, 0.96, and 0.82. The channels therefore play a dual role: they are higher-resolution features for the TFT, and, in the absence of human ground truth, cross-model agreement on them serves as a defensible validation signal for the extraction [Gilardi et al., 2023, Zheng et al., 2023].

Tone versus price impact. A tone classifier and a price-reasoning LLM are not a coarse and a fine version of the same measurement; they measure different constructs (Section 4.3.8, Section 5.2). FinBERT reports the surface financial tone of an article, while the LLM performs the asset-specific causal step, from a supply threat to a higher crude price, that a tone classifier cannot. The geopolitical sign-flip is the empirical demonstration. The lesson generalizes beyond this thesis: for an asset-specific task, sentiment must be defined relative to the asset and its causal mechanism, not as generic positive-or-negative tone, and a study that conflates the two can reach opposite conclusions depending on which it happens to use.

LLM-as-filter for noisy news corpora. Phase 1’s inclusion rule was a brittle regex heuristic on the article body (Section 3.4.1); Phase 2 replaced it with the LLM’s own judgment of whether an article is materially relevant (`usable`), refined to `usable_strict` by additionally requiring at least a non-zero channel score (Section 4.3.3). The LLM filter is semantically grounded where the regex is lexical, and the comparison of the two (Section 4.3.3) shows where each fails. For web-scraped news, where relevance is a matter of meaning rather than keywords, a semantically grounded filter is a practical and transferable alternative to keyword heuristics.

Phase-based progression as a reproducible template. The two-phase design is itself a contribution. A simple, interpretable baseline (FinBERT features with OLS) answers the research questions with classical statistics whose assumptions are transparent; a richer extraction with a more expressive model (LLM channels with a TFT) then both corroborates the baseline and enriches the analytical vocabulary. Neither phase is

discarded: the second does not supersede the first but triangulates it, and each one has limitations of its own. This progression is a reusable template for empirical news-impact studies that want both interpretability and expressiveness without sacrificing either.

5.4 Limitations

Several limitations qualify the conclusions and delimit their scope.

Inter-model calibration is weaker than human ground truth. The LLM features are validated by cross-model agreement (Haiku versus GPT-5.5, Section 4.3.4) rather than against expert human annotation, which was outside the resource scope (Section 3.7). Cross-family agreement is informative, two models built by different developers converging on the same reading is evidence of something real, but it is not a gold standard: both models can share the same blind spot, and agreement does not guarantee correctness. The calibration sample is also small ($n = 30$) by design, enough to surface systematic disagreements but not to estimate population agreement precisely.

Regime extrapolation on price_range. The most concrete modeling limitation is the failure of TFT v2 on `price_range` in the war regime (Section 4.3.7.3). Trained only on the pre-war period, the model reverts toward the historical mean when asked to predict the high-volatility war regime it never saw, underperforming even a naive persistence baseline. This is a clean illustration of a model informationally constrained by its training window, and a warning that the aggregate liquidity results (`log_volume`, `amihud`), which transfer better across the regime boundary, should not be assumed to generalize to every target.

Construct dependence of the sentiment findings. As Section 5.2 makes explicit, the directional conclusions depend on how sentiment is defined (tone versus price impact) and on which quantity is estimated (marginal coefficient versus group mean). This is a strength when stated carefully, but a limitation for anyone seeking a single unconditional answer: there is no framing-free statement of the asymmetry, and any restatement must carry its definitions with it.

Corpus composition. The news corpus is drawn from GDELT (Section 3.1.4), a broad machine-monitored aggregator that skews toward retail-facing web and print coverage rather than the institutional-grade feeds (trading terminals, wire services) that professional oil traders actually consume. The sentiment the models extract is therefore the sentiment of publicly available news, which may lead, lag, be biased, or simply differ from the information set that moves institutional order flow.

Non-stationarity and single-commodity scope. The dataset spans roughly 24 months and contains a structural break (the Iran war onset, Section 4.3.1) that makes the series non-stationary across the train and test boundary; the war-regime results are on a single

episode and should be read as suggestive rather than established. The study is also confined to a single instrument, WTI crude oil futures. The channel decomposition and the risk-and-salience reading of the liquidity response are plausibly specific to a commodity with WTI’s geopolitical supply structure, and neither the lag nor the asymmetry findings can be assumed to carry to other commodities or asset classes without replication.

5.5 Future work

The limitations above point to several concrete extensions, most of them small enough to be tractable.

Small-scale human calibration. The clearest methodological gap is the absence of human ground truth (Section 5.4). A tractable next step is not full-scale annotation but a small expert-annotated anchor set, a modest 200 articles scored by an oil-market professional, against which both the LLM extraction and the cross-model agreement can be benchmarked. This would turn the inter-model calibration from a standalone signal into a calibrated one, without the cost of annotating the full corpus.

Separating tone and price sentiment as parallel features. The construct-dependence finding (Section 5.2) suggests an experiment rather than only a caveat: carry both the FinBERT tone score and the LLM price-impact score as separate features and let the model weigh them. For a liquidity target in particular, where salience and risk drive volume more than price direction, the tone signal may carry information the price signal does not, and their disagreement (the geopolitical sign-flip) may itself be a useful feature marking high-uncertainty news.

Multi-regime training. The `price_range` extrapolation failure (Section 5.4) is a training-window problem, not a model-capacity one. As more regime episodes accumulate in the data, retraining across multiple structural breaks, or explicitly conditioning on a regime indicator, would test whether the volatility target becomes learnable once the model has seen more than one regime.

Cross-commodity replication. The risk-and-salience reading of the response (Section 5.2) is a hypothesis about oil specifically, where geopolitical supply risk is price-bullish. Replicating the two-phase design on commodities with different supply and demand structures (natural gas, gold, copper) would test whether the same risk-premium mechanism governs their news-liquidity response or whether it is specific to WTI’s geopolitical exposure. Cross-commodity spillovers, oil news moving gas or gold liquidity, are a further and larger question left open here.

Richer entity representation. The entity flags are a binary vocabulary of 71 canonical actors (Section 4.3.2). Replacing them with domain-aware entity embeddings, learned

representations that place, for instance, Oman and Qatar near each other as Hormuz-adjacent producers, would let the model generalize across related actors rather than treating each as an independent indicator, and could sharpen the risk-and-salience signal the current flags only approximate.

6. Conclusion

This thesis set out to understand and model how news propagates into the liquidity of WTI crude oil futures at hourly resolution, along two axes: the temporal lag of the response (RQ1) and its directional asymmetry (RQ2). It pursued the questions through a two-phase design, a regression-based Phase 1 on FinBERT sentiment and a deep-learning Phase 2 on LLM-extracted, channel-decomposed features.

This chapter states the contributions, summarizes the findings, and closes.

6.1 Summary of contributions

The thesis makes three empirical and four methodological contributions.

Empirically:

- **An hourly lag structure for the news-liquidity response.** The response peaks in the +6 to +12 hour window, established by the Phase 1 lag OLS and independently corroborated by the Phase 2 TFT through both its per-horizon error curve and its direction-conditioned attention. This is structure that daily-frequency work cannot observe.
- **A precise account of the directional asymmetry.** There is a robust bearish over bullish asymmetry in the marginal sensitivity of volume to negative-tone news (Phase 1). It does not appear as a difference in the level of predicted volume (Phase 2), because the model organizes its predictions around risk and salience rather than bullish or bearish price direction. The contribution is as much the precision as the result: stating the asymmetry correctly requires naming the sentiment construct and the estimand it refers to.
- **A news-representation finding.** Title-only and title-plus-body FinBERT sentiment disagree on 41.6% of articles, with titles leaning more bullish than the bodies they head, evidence that the choice of news representation is itself a modeling decision.

Methodologically (developed in Section 5.3):

- **Channel decomposition** as a higher-resolution feature set and, through cross-model agreement, a moderately defensible validation substrate in the absence of

human ground truth.

- **The tone-versus-price distinction**, showing that a sentiment measure is defined by its construct, and that a tone classifier and a price-reasoning LLM diverge on exactly the geopolitical news that drives volume.
- **LLM-as-filter** as a semantically grounded alternative to regex heuristics for noisy web-scraped news corpora.
- **A two-phase template** that pairs an interpretable statistical baseline with an expressive deep-learning model that corroborates rather than replaces it.

6.2 Summary of findings

On RQ1, news affects WTI liquidity most strongly not in the same hour but over a +6 to +12 hour window. Two independent methods agree: the Phase 1 lag OLS peaks at +6 hours, and the Phase 2 TFT, trained and evaluated separately, places its strongest persistence-relative gain in the same window and attends to −6 hours for bearish news. The liquidity response to news is gradual, not instantaneous.

On RQ2, the answer depends on how the question is posed. There is a robust bearish over bullish asymmetry in the marginal sensitivity of volume to negative-tone news, and this is the thesis’ primary answer. It does not survive as a difference in the average level of predicted volume, because the model organizes its forecasts around the risk and salience a news item carries rather than its price direction, and because a tone-based and a price-based sentiment measure diverge on exactly the geopolitical news that moves the market. WTI liquidity responds to how much a piece of news raises supply uncertainty more than to whether it reads as nominally good or bad.

As a forecasting exercise, TFT v2 cut `log_volume` error over a persistence baseline by 46 to 71 percent across horizons and `amihud` by 43 to 45 percent, with VIX, the supply and demand channels, and geopolitical entity flags as its leading features; it failed only on `price_range` in the unseen war regime, a clean case of regime extrapolation. Alongside the two headline answers, three supporting findings shaped the research development: a 41.6 percent title-versus-body sentiment disagreement, a composite-sentiment calibration failure between model families that motivated the channel decomposition, and a semantic-versus-lexical filter comparison that motivated the `usable_strict` rule.

6.3 Closing remarks

This thesis began from a simple observation, that news moves markets, and asked a more precise question of it: at what lag, and with what asymmetry, does news move the liquidity

of WTI crude oil at hourly resolution. The answer that emerged is itself twofold. The empirical answer is that the response is gradual and peaks over a +6 to +12 hour window, and that its directional asymmetry is real but must be stated with care, because it lives in the sensitivity of volume to news, not in the average volume of nominally good or bad hours. The methodological answer is that reaching this precision required treating sentiment not as a single given quantity but as a construct to be defined, decomposed, and validated, which is where the channel decomposition, the LLM extraction, and the two-phase design earn their place.

If one idea carries beyond this study, it is that for a commodity like crude oil the liquidity response to news is organized around risk rather than direction. The market does not trade more because the news is bad; it trades more because the news raises uncertainty about supply, and in the oil market that uncertainty is often, against intuition, bullish for price. Recognizing this dissolves the apparent tension between a tone-based and a price-based reading of the same events, and reframes what a news-driven liquidity model should be built to detect.

The work is bounded: one commodity, two years, one structural break, and a validation resting on model agreement rather than human judgment. But within those bounds it offers both a concrete empirical result and a reusable way of reaching it, and the extensions of Section 5.5 are, for the most part, small steps rather than new programs. The propagation of news into market liquidity is a question of resolution as much as of theory, and at hourly resolution it shows more structure, and more nuance, than a daily view can.

A. Extraction prompt

This appendix reproduces the Haiku v2 extraction interface: the system prompt, the `extract_article_features` tool schema passed to the tool-use API, and the request configuration. The conceptual field-level schema and its revision history are in Appendix B. The source of record is `03_src/nlp/llm_features.py`. Dashes in the reproduced prompt are normalised to ASCII for typesetting.

A.1 System prompt. The following text is passed as the cached system prompt.

```
You are a financial analyst specializing in WTI crude oil markets
. Your task is to analyze news articles and extract structured
features for use in a quantitative liquidity impact model.

First, determine whether the article contains substantive WTI-
relevant content (usable=true). Set usable=false -- and return
only that field -- for: paywall placeholders, Cloudflare blocks
, cookie notices, or articles that are clearly not about oil
markets, macro, or geopolitics relevant to oil.

When usable=true, extract the remaining fields:
- sentiment_score: NET directional impact on WTI price. -1.0 =
  strongly bearish, +1.0 = strongly bullish. Assess the actual
  content, not the headline tone.
- supply_impact: Direction of impact on WTI supply. Score the
  article's implication for the quantity of oil available to the
  market.
+1.0 = strong supply increase (OPEC adds barrels, sanctions
  lifted on a major producer, new pipeline online, Venezuela
  ramping up).
+0.5 = moderate supply increase (single-country production
  gains, inventory builds).
0.0 = no supply implication.
-0.5 = moderate supply decrease (refinery outages, minor OPEC
  cuts).
-1.0 = strong supply decrease (major OPEC cuts, infrastructure
  attacks, embargoes, refinery fires affecting multiple
  facilities).
```


- demand_impact: Direction of impact on WTI demand. Score the article's implication for oil consumption.
 - +1.0 = strong demand increase (China stimulus, Fed pivot to easing, growth surprise, summer driving acceleration).
 - +0.5 = moderate demand increase (regional growth indicators).
 - 0.0 = no demand implication.
 - 0.5 = moderate demand decrease (mild slowdown signals).
 - 1.0 = strong demand decrease (recession confirmed, Fed strongly hawkish, major economy lockdowns, demand destruction events).
- risk_premium: Change in geopolitical or operational risk premium for oil. Score the direction of risk change, not the absolute level.
 - +1.0 = major escalation (military strike on oil infrastructure, war, Strait of Hormuz incident, major sanctions imposed).
 - +0.5 = moderate escalation (rising tensions, threats, troop movements).
 - 0.0 = no change in risk environment.
 - 0.5 = moderate de-escalation (talks resuming, partial sanctions lifted).
 - 1.0 = major de-escalation (ceasefire signed, diplomatic breakthrough, full sanctions removal).
- magnitude: how market-moving is this event. 0.0 = negligible, 1.0 = historic. Most articles score 0.1-0.4. Major OPEC cut = 0.9. Geopolitical crisis = 1.0.
- event_type: 1-3 categories ordered by salience from: geopolitical, supply, demand, macro, technical, other. EIA inventory articles classify as supply or macro -- do not use "inventory".
- entities: specific named actors central to the article (not incidentally mentioned). Include countries, organizations, oil companies, and key officials. Use the names exactly as they appear in the article.
- certainty: how confirmed is the information. 0.0 = rumor, 0.5 = analyst forecast, 0.9 = official announcement.
- time_horizon: immediate (hours to 1 day), short_term (days to weeks), or structural (months-plus or permanent themes such as energy transition).

The three impact fields (supply_impact, demand_impact, risk_premium) are intentionally orthogonal. An article can

affect multiple channels independently. Score each based on what the article actually says about that specific channel. If the article has no clear implication for a channel, score it 0.0.

A.2 Tool schema. The model is forced to call a single tool, `extract_article_features`, whose `input_schema` is reproduced below. Only `usable` is required; every other field is omitted when `usable=false`.

```
{
  "name": "extract_article_features",
  "description": "Extract structured WTI-relevant features from a
                  news article
                  for use in a quantitative liquidity impact
                  model.",
  "input_schema": {
    "type": "object",
    "properties": {
      "usable": { "type": "boolean" },
      "sentiment_score": { "type": "number" },
      "supply_impact": { "type": "number", "minimum": -1.0, "
                        maximum": 1.0 },
      "demand_impact": { "type": "number", "minimum": -1.0, "
                        maximum": 1.0 },
      "risk_premium": { "type": "number", "minimum": -1.0, "
                       maximum": 1.0 },
      "magnitude": { "type": "number" },
      "event_type": { "type": "array",
                      "items": { "type": "string",
                                "enum": ["geopolitical", "supply", "
                                         demand",
                                         "macro", "technical", "other"]
                                },
                      "minItems": 1, "maxItems": 3 },
      "entities": { "type": "array", "items": { "type": "
                string" } },
      "certainty": { "type": "number" },
      "time_horizon": { "type": "string",
                       "enum": ["immediate", "short_term", "
                                structural" ]
    },
    "required": ["usable"]
  }
}
```

```
}  
}
```

Each numeric field carries a per-field **description** in the live schema that restates the scoring anchors of the system prompt; those are elided here for brevity and are recoverable from the source file. Note that `time_horizon` is a single categorical label, not an array (only `event_type` is multi-valued).

A.3 Request configuration. Requests are submitted through the Anthropic Batches API, one request per article, with the following parameters:

- `model`: `claude-haiku-4-5`
- `max_tokens`: 500
- `system`: the A.1 prompt, sent with `cache_control` ephemeral so the prompt is cached across the batch
- `tools`: the A.2 tool; `tool_choice` set to force `extract_article_features`
- user message: the title and body, where the body is UTF-8 sanitised and truncated to the first 1,500 characters (title only when no body was retrieved)

B. LLM extraction schema (v1 vs v2)

The Haiku extraction schema was revised once during Phase 2, from Schema v1 (used for TFT v1, Section 4.3.5) to Schema v2 (the canonical extraction used for all reported Phase 2 results, Section 4.3.6). The revision was the direct response to the inter-model calibration finding of Section 4.3.4: the composite `sentiment_score` was unreliable on high-magnitude geopolitical events because it conflated event valence with directional price impact.

Field-by-field comparison.

Table B.1: Schema v1 versus v2, field by field.

Field	Schema v1	Schema v2
<code>sentiment_score</code>	continuous $[-1, +1]$	continuous $[-1, +1]$ (retained)
<code>supply_impact</code>	n/a	continuous $[-1, +1]$ (new)
<code>demand_impact</code>	n/a	continuous $[-1, +1]$ (new)
<code>risk_premium</code>	n/a	continuous $[-1, +1]$ (new)
<code>magnitude</code>	$[0, 1]$	$[0, 1]$
<code>certainty</code>	$[0, 1]$	$[0, 1]$
<code>event_type</code>	single categorical label	array of one to three labels
<code>time_horizon</code>	categorical label	categorical label
<code>price_direction</code>	categorical label	dropped
<code>entities</code>	list of canonical actors	list of canonical actors
<code>usable</code>	implicit (prompt only)	required boolean field

Changes and rationale.

1. **Three orthogonal economic channels added** (`supply_impact`, `demand_impact`, `risk_premium`, each on $[-1, +1]$). This decomposes the single sentiment judgment into its separable economic drivers, mirroring the standard structural decomposition of oil-price shocks into supply, aggregate-demand, and precautionary (risk) components [Kilian, 2009]. Re-calibration lifted the composite `sentiment_score` cross-model correlation from 0.39 to 0.88 and gave the channels correlations of 0.94 (supply), 0.96 (demand), and 0.82 (risk) (Section 4.3.6).
2. **usable promoted to a required field** (from an implicit prompt instruction), enabling the LLM-based filter and the derived `usable_strict` variant (Section 4.3.3).

3. **event_type** changed from a single label to an array of one to three salience-ordered labels, so an article that is jointly geopolitical and supply-related (e.g. Iran sanctions) need not be forced into a single category.
4. **price_direction** dropped: it agreed with the sign of **sentiment_score** on all but roughly 16 of 12,024 articles and carried about 0.008 feature importance in TFT v1, and the three channels provide a richer directional substrate.
5. **sentiment_score** retained (not replaced by the channels), for continuity with the headline-bias experiment (Section 4.2.2) and as a hedge against individually noisy channels.

TFT v1 (Section 4.3.5) was trained on Schema v1; TFT v2 (Section 4.3.7) and all reported Phase 2 results use Schema v2.

C. TFT v2 ablation and hyperparameters

The reported TFT v2 (variant v2.2) was selected through a three-variant ablation, all on the 60/20/20 split with `usable_strict=1`, dropout 0.15, and early-stopping patience 10; only the design axis changes between rows.

Table C.1: Three-variant TFT v2 ablation.

Variant	Categorical encoding	Targets $\times$ horizons	Entity flags	Val loss	MAE@1h	Feature #
v2.0	int-encoded	<code>log_volume</code> $\times$ 1h	no	0.257	0.455	sentiment
v2.1	proper categoricals	<code>log_volume</code> $\times$ 1h	no	0.258	0.424	demand_i
v2.2 (canonical)	proper categoricals	$3 \times [1, 3, 6, 12]$	71	0.427*	0.585	vix 0.188

* v2.2's `val_loss` (`MultiLoss` over three targets) is not comparable to the single-target v2.0/v2.1. The key finding is that proper categorical encoding (v2.0 to v2.1) is what unlocks the channels' predictive role: `demand_impact` jumps to 0.485. The full per-run record is in `05_reports/v2_training_results.md`.

Canonical TFT v2 hyperparameters. Hidden size 32, attention head size 4, hidden-continuous size 16, dropout 0.15, learning rate 1e-3 (Adam, on-plateau patience 3), gradient clip 0.1, early-stopping patience 10, seed 42; 298,329 trainable parameters; encoder length 48h; targets [`log_volume`, `amihud`, `price_range`]; horizons [1, 3, 6, 12]; split 60/20/20 (TRAIN_END 6739, VAL 6787–9014, TEST 9062–11232).

D. Canonical entity list

The 71 canonical entities used as TFT v2 entity flags (from `03_src/nlp/llm_features.py`), each mapped from raw article mentions via the normalization of Section 4.3.2 (alias mappings omitted here for brevity):

US, Iran, Russia, China, Israel, India, Saudi Arabia, Ukraine, Venezuela, Canada, Iraq, Nigeria, UAE, Kazakhstan, Qatar, Oman, Japan, Kuwait, Pakistan, Libya, Mexico, Azerbaijan, Yemen, Lebanon, Brazil, South Korea, Guyana, UK, Algeria, Germany, Australia, Hungary, Egypt, Türkiye, Strait of Hormuz, Middle East, Gaza, Red Sea, Persian Gulf, Gulf of Mexico, Permian Basin, Europe, Asia, Trump, Maduro, Putin, Powell, Bessent, Biden, OPEC+, OPEC, Fed, EU, EIA, IEA, API, ECB, Hamas, Hezbollah, Houthis, Saudi Aramco, Chevron, Shell, BP, ExxonMobil, Rosneft, Lukoil, TotalEnergies, WTI, Brent, S&P 500.

E. Inter-model calibration comparison

This appendix gives the article-by-article record behind the inter-model calibration of Sections 3.7 and 4.3.4. The same 30-article stratified sample was scored independently by Claude Haiku 4.5 (the extraction model, denoted H) and by GPT-5.5 (an independent reference from a different model family, accessed through the ChatGPT interface, denoted G), using the identical Schema v2 prompt and tool of Appendix A. Two independently trained models agreeing is a defensible, if weaker than human, validation signal; disagreement flags genuine ambiguity in the article.

Aggregate agreement (Schema v2). The two models agree on the binary `usable` flag for 27 of 30 articles; the three disagreements (IDs 13331, 10308, 10211) are borderline single-company or forecast pieces that Haiku filtered out and GPT retained. On the 14 articles both models judged usable, the composite `sentiment_score` reaches a Pearson correlation of 0.88, with a single sign disagreement (1/14, using a 0.1 neutral band, so only scores with magnitude above 0.1 count as directional). The three channels reach Pearson correlations of 0.94 (supply), 0.96 (demand), and 0.82 (risk). For contrast, Schema v1, which scored only the composite, gave a sentiment correlation of 0.39 with 4 sign disagreements out of 13; the channel decomposition of Schema v2 is what lifted agreement (Section 4.3.6).

Usability and composite sentiment (all 30 articles).

Table E.1: Per-article usability and composite sentiment, Haiku (H) versus GPT (G).

ID	Title	Us. H	Us. G	Sent. H	Sent. G
11630	Top Stock Movers Now: Vertex, DoorDash, Con...	×	×		
16558	Oil prices fall as analysts forecast an increa...	✓	✓	-0.60	-0.68
18615	WTI Forecast: Retreats from 7-month top, h...	✓	✓	+0.70	+0.74
13331	Crude Oil Weekly Forecast 10/08: Tempting S...	×	✓		-0.42
13772	Focus on Nigeria Non-oil Revenue Potential	×	×		

Table E.1 (continued)

ID	Title	Us. H	Us. G	Sent. H	Sent. G
4108	Oil up on Opec+ meeting, summer driving seaso...	✓	✓	+0.60	+0.81
8677	Top 5 things to watch in markets in the week a...	✓	✓	+0.00	-0.16
3151	Oil recovers after surprise drawdown in US sto...	✓	✓	+0.30	+0.37
3779	Global supply chains now in Goldilocks Zone...	✓	✓	+0.40	+0.21
6068	Oil climbs with U.S. Federal Reserve pivot	✓	✓	+0.60	+0.58
3335	Gran Tierra Energy Inc. Announces First Quart...	✓	✓	+0.30	-0.06
10308	Vermilion Energy Inc. Announces Results for t...	×	✓		-0.02
7834	Is Occidental Petroleum Corporation (OXY) th...	✓	✓	+0.20	+0.17
7188	ALPS Advisors Inc. Expands Stake in En- Link Mi...	×	×		
10989	Why Occidental Petroleum Stock Plunged 10% To...	✓	✓	-0.30	+0.09
10799	Is Atmos Energy Corp. (ATO) The Best Defens...	×	×		
13211	Baytex Energy Corp.: Baytex Delivers Solid Se...	✓	✓	+0.30	-0.12
20149	Why Halliburton (HAL) Stock Is Trading Up To...	✓	✓	+0.70	+0.86
10211	Is Chevron Corporation (CVX) The Best Crude...	×	✓		-0.35
6161	1 Growth Stock Down 18% to Buy Right Now	×	×		
8882	Petroleum prices increase ahead of Christ- mas	×	×		
6160	Stock market today: US stocks slip as traders...	×	×		
13217	The Latest: Market slumps, Gaza blood- bath an...	×	×		
11929	Oil Prices Rise Amid US-China Trade Opti- mism...	×	×		
16355	Iran Warns US Intervention Would Destabi- lize R...	✓	✓	+0.30	+0.25
21358	How China built an energy fortress to survive o...	×	×		
21692	Global oil prices surge above \$111 per bar- rel...	✓	✓	+0.70	+0.85

Table E.1 (continued)

ID	Title	Us. H	Us. G	Sent. H	Sent. G
1636	Gasoline Market 2024-2032: Industry Re- port...	×	×		
19074	Governments Move to Shield Economies as Oil Ju...	×	×		
18	Oil jumps 1% in New Year after US forces repe...	×	×		

Channel scores (14 articles both models judged usable).

Table E.2: Per-article channel scores on the both-usable articles, H versus G.

ID	Supply H	Supply G	Demand H	Demand G	Risk H	Risk G
16558	+0.70	+0.72	-0.40	-0.34	-0.30	+0.18
18615	+0.00	-0.12	+0.00	+0.00	+0.60	+0.78
4108	-0.50	-0.72	+0.50	+0.48	+0.00	+0.00
8677	+0.00	+0.00	+0.00	-0.22	+0.00	+0.08
3151	-0.50	-0.44	+0.00	-0.12	+0.00	+0.00
3779	+0.00	+0.00	+0.50	+0.34	+0.00	+0.00
6068	+0.00	-0.08	+0.70	+0.52	+0.00	+0.00
3335	+0.40	+0.29	+0.00	+0.00	+0.00	+0.00
7834	+0.00	-0.12	+0.00	+0.00	+0.00	+0.44
10989	+0.00	+0.06	+0.00	+0.00	+0.00	+0.00
13211	+0.40	+0.33	+0.00	+0.00	+0.00	+0.00
20149	+0.00	-0.48	+0.00	+0.00	+0.50	+0.91
16355	+0.00	+0.00	+0.00	+0.00	+0.50	+0.45
21692	-0.60	-0.75	+0.00	+0.00	+0.40	+0.60

The residual disagreement concentrates in the risk channel ($r = 0.82$, the lowest of the three): on IDs 16558, 20149, and 7834 the two models weight the geopolitical or operational risk implication quite differently, whereas supply and demand track closely. This is consistent with risk being the hardest channel to score, noted in Section 4.3.6.

F. Extended metrics tables

This appendix reports the full TFT v2 test-set metrics summarised in Section 4.3.7.3. For each of the three targets it gives mean absolute error (MAE) and root mean squared error (RMSE) at each forecast horizon against the persistence baseline ($\hat{y}_{t+k} = y_t$, the last observed value carried forward), broken out by the full test set and by the pre-war and war regimes (split at the war onset, decoder time index 10056). “Reduction” is the percentage by which the TFT lowers MAE relative to persistence; a positive value means the model beats persistence. The test set has 2,159 forecast origins; the pre-war and war row counts differ slightly across horizons because a fixed origin’s $+k$ -step target crosses the war boundary at different k .

Table F.1: `log_volume` extended metrics.

Horizon	Slice	n	TFT MAE	Persist. MAE	Reduction	TFT RMSE	Persist. RMSE
+1h	overall	2159	0.585	1.076	+46%	0.915	2.334
+1h	pre-war	994	0.536	1.075	+50%	0.863	2.233
+1h	war	1165	0.628	1.076	+42%	0.957	2.417
+3h	overall	2159	0.577	1.452	+60%	0.895	2.734
+3h	pre-war	992	0.532	1.480	+64%	0.855	2.664
+3h	war	1167	0.614	1.428	+57%	0.928	2.793
+6h	overall	2159	0.602	1.820	+67%	0.932	2.838
+6h	pre-war	989	0.543	1.935	+72%	0.865	2.841
+6h	war	1170	0.652	1.724	+62%	0.985	2.835
+12h	overall	2159	0.631	2.174	+71%	0.958	3.062
+12h	pre-war	983	0.555	2.319	+76%	0.874	3.111
+12h	war	1176	0.695	2.053	+66%	1.024	3.021

Reading. Three patterns stand out. On `log_volume`, the primary liquidity target, the TFT beats persistence at every horizon and in both regimes, and the margin widens with horizon (from 46% at +1h to 71% at +12h overall), consistent with the delayed-propagation reading of RQ1 and stable across the war boundary. On `amihud`, the TFT also beats persistence throughout (42 to 46% overall); absolute errors are on the order of 10^{-4} and are dominated by war-regime spikes. On `price_range`, the picture splits by regime: the TFT beats persistence in the pre-war window (reduction rising to +43% at +12h) but underperforms it overall and in the war regime (negative reductions). Trained only on pre-war data, the model reverts toward the historical mean and cannot reach the war-regime volatility spikes (actual `price_range` up to 14.4 against a predicted ceiling near 1.5), the regime-extrapolation failure discussed in Section 4.3.7.3 and Section 5.4.

Table F.2: amihud extended metrics (MAE and RMSE in units of 10^{-3}).

Horizon	Slice	n	TFT MAE	Persist. MAE	Reduction	TFT RMSE	Persist. RMSE
+1h	overall	2159	0.23	0.41	+42%	2.73	3.86
+1h	pre-war	994	0.10	0.15	+31%	0.60	0.87
+1h	war	1165	0.35	0.62	+45%	3.67	5.20
+3h	overall	2159	0.23	0.41	+44%	2.72	3.86
+3h	pre-war	992	0.10	0.15	+36%	0.59	0.87
+3h	war	1167	0.34	0.62	+46%	3.65	5.19
+6h	overall	2159	0.22	0.41	+45%	2.72	3.86
+6h	pre-war	989	0.09	0.15	+37%	0.59	0.87
+6h	war	1170	0.33	0.62	+46%	3.65	5.19
+12h	overall	2159	0.22	0.41	+45%	2.72	3.86
+12h	pre-war	983	0.09	0.15	+38%	0.60	0.87
+12h	war	1176	0.33	0.62	+46%	3.64	5.17

Table F.3: price_range extended metrics.

Horizon	Slice	n	TFT MAE	Persist. MAE	Reduction	TFT RMSE	Persist. RMSE
+1h	overall	2159	0.718	0.495	-45%	1.287	0.981
+1h	pre-war	994	0.154	0.180	+14%	0.256	0.278
+1h	war	1165	1.200	0.764	-57%	1.736	1.310
+3h	overall	2159	0.719	0.578	-24%	1.286	1.107
+3h	pre-war	992	0.155	0.220	+30%	0.257	0.329
+3h	war	1167	1.199	0.883	-36%	1.733	1.475
+6h	overall	2159	0.721	0.630	-14%	1.289	1.194
+6h	pre-war	989	0.156	0.233	+33%	0.256	0.353
+6h	war	1170	1.198	0.966	-24%	1.734	1.589
+12h	overall	2159	0.724	0.701	-3%	1.294	1.252
+12h	pre-war	983	0.157	0.278	+43%	0.257	0.390
+12h	war	1176	1.198	1.055	-14%	1.737	1.658

G. Reproducibility statement

The Phase 2 training was run on Google Colab (T4 GPU); the LLM extraction and the Phase 1 analyses were run locally (Apple Silicon, MPS backend). Software with a paper of record is cited; the rest are attributed here by name, version, and source.

Software libraries. Version constraints are pinned in the training notebook 02_notebooks/13_tft_v2_training_colab.ipynb.

Table G.1: Software libraries and versions.

Library	Version	Role
Python	3.x	runtime
numpy	< 2.0	numerics
pandas	n/a	data processing and hourly aggregation
PyTorch (torch)	$\geq 2.3, < 2.4$	deep-learning backend
Lightning (lightning)	$\geq 2.2, < 2.4$	training loop
pytorch-forecasting	$\geq 1.0, < 1.2$	TFT implementation
HuggingFace transformers	n/a	FinBERT inference [Wolf et al., 2020]
yfinance	n/a	market-data retrieval
beautifulsoup4	n/a	article-body HTML parsing
scipy	n/a	statistical tests (Welch t-test, correlations)
sqlite3 (stdlib)	n/a	database

Models and APIs.

- Claude Haiku 4.5 (Anthropic), via the tool-use API, for Phase 2 feature extraction (Section 3.3).
- GPT-5.5 (OpenAI), via the ChatGPT interface, as the inter-model calibration reference (Section 3.7).

Data sources.

- GDELT 2.0 Document API [Leetaru and Schrodt, 2013], for news article metadata (<https://www.gdeltproject.org>).
- U.S. Energy Information Administration (EIA) v2 API, weekly crude inventory series WCRSTUS1 (<https://www.eia.gov/opendata/>).
- yfinance / Yahoo Finance, hourly OHLCV for CL=F, DX=Y.NYB, and ^VIX.

Determinism. The reported TFT v2 run fixes the random seed to 42 with deterministic algorithms; its exact configuration is stored in `run_metadata.json`. Exact pinned library versions and data-source access dates are to be recorded from the environment at finalisation.

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